

Spectrum-Matched Quantum-Assisted PINNs: A Constructive Design Methodology for Problem-Specific Variational Circuits, Instrumented by Explainable-AI Diagnostics

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Abstract

A classical physics-informed neural network (PINN) fails on exactly one thing, reliably and provably: it learns low frequencies fast and high frequencies almost not at all (spectral bias), because its neural tangent kernel (NTK) eigenspectrum decays with Fourier-mode index. A data-re-uploading variational quantum circuit is not a black-box speedup — it is a Fourier feature map whose accessible frequency set is exactly and prescribably determined by the eigenvalues of its data-encoding Hamiltonians. This paper joins those two facts into a constructive design rule, Spectrum-Matched Circuit Design (SMCD, Section 3): given a PDE, we compute the target spectral support analytically from the operator symbol and forcing/boundary data, then construct the encoding generators, re-uploading depth, and qubit count so the circuit’s realized frequency set covers it — no architecture search. We use six explainable-AI instruments, run identically on matched classical and hybrid models, to test that mechanism rather than merely report an accuracy delta. Against twelve predictions committed before any experiment ran and adjudicated by a script, at two seeds per configuration the hybrid model does not outperform the classical baselines: a classical model given the same frequencies matches or beats it on every problem, and the predicted NTK flattening is reversed where measurable. The design rule itself holds up: the circuit’s output stays within its designed frequency set, 99.9% of its per-frequency improvement on the Poisson problem lies inside that set, and on the heat equation it reaches $12\times$ lower error than a same-size random circuit, though two seeds cannot establish significance.

1 Introduction

A classical physics-informed neural network (PINN) learns low-frequency solution components fast and high-frequency components almost not at all — spectral bias — and its neural tangent kernel (NTK) eigenspectrum decays with Fourier-mode index, so residual and boundary loss terms converge at characteristically different rates. Separately, a data-re-uploading variational quantum circuit is known to realize a truncated Fourier series whose accessible frequency set is exactly determined by its data-encoding Hamiltonians’ eigenvalues [10]. These two facts have not previously been joined into a constructive design rule: prior hybrid-QPINN work [2, 6, 11, 14] reports empirical accuracy deltas on hand-picked circuit sizes, without a principled way to choose the circuit *given* the PDE.

This paper makes that connection. Given a PDE, we compute its target spectral support analytically from the operator’s Fourier symbol and forcing/boundary data, then construct the quantum

circuit’s encoding generators, re-uploading depth, and qubit count so its realized frequency set covers that support — no architecture search, no hyperparameter sweep over circuit size. We call this Spectrum-Matched Circuit Design (SMCD, Section 3). We then use six explainable-AI instruments, run identically for matched classical and hybrid models at the same training checkpoints, as the measurement instrument that attributes any accuracy delta to the quantum layer specifically — not as after-the-fact decoration on a result already claimed by other means.

Contributions.

- C1 — SMCD, a constructive PDE $\rightarrow$ circuit map.** Given a linear operator’s symbol and forcing/boundary spectra, output encoding-generator eigenvalues, re-uploading depth, qubit count, and entangler pattern, deterministically. Falsifiable: a spectrum-matched circuit must beat a size-matched random-frequency circuit on the same PDE, or the design step is not earning its keep.
- C2 — Quantum-NTK block decomposition.** $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q$ for disjoint-parameter hybrid models (Section 3.5), with the quantum block’s eigenvalue decay flattened over the encoded band relative to the classical block’s power-law decay.
- C3 — Spectral-bias relief as mechanism, not just outcome.** Per-frequency error trajectories must show the hybrid model’s high- k modes converging at rates comparable to its low- k modes, with the improvement band coinciding with the encoded frequency set — not merely an aggregate accuracy improvement.
- C4 — A negative-result map with a decision rule.** An explicit characterization of PDE classes where hybridization is useless or harmful, with the mathematical reason (smoothing operators remove high-frequency content, so a high-frequency-rich circuit adds parameters and barren-plateau risk for no benefit).
- C5 — An XAI protocol for hybrid physics-informed models.** A reusable, model-agnostic instrument set with published metric definitions, run identically across classical and quantum families rather than as a post hoc, model-specific explanation.

The honest headline this paper reports, whatever the specific numbers turn out to be (Section 5–6): the quantum layer should help on high-wavenumber and multiscale problems where its encoded band matches the PDE’s spectral support, do nothing on smooth diffusive problems whose target spectrum SMCD predicts to be low-frequency-only *before any training*, and hurt when the circuit’s spectrum is mismatched to the problem. A design rule plus a stated “when not to go quantum” recommendation, adjudicated against pre-registered falsifiers (Section 5), is the contribution — not a cherry-picked win.

Assignment mapping (T5.9 self-review).

Requirement	Where it is satisfied
Solve a set of PDEs with classical PINNs and QAPINNs	Section 3 (4 PDE instances $\times$ 6 problem labels, §3.6), Section 3 (7 model families), one training harness for both
XAI techniques/metrics showing <i>how</i> learning happens in each	Section 4 (6-instrument suite), run on both families at matched checkpoints, results in Section 5
Demonstrate how the VQC affects the classical PINN learning process	Section 5 (NTK block decay, PR-7; per-frequency convergence, PR-8; matched- capacity ablation, PR-1–3), Section 6 (negative controls, PR-4, PR-11)
Methodology to construct a problem-specific circuit + architecture, with mathematical basis	Section 3 (SMCD, Propositions 1–4, Algorithm 1)
Technical report, 5–15 pp	This document, <code>paper/main.pdf</code> (15 pp as submitted, repository tag <code>v1.0-submission</code> ; this revision adds figures and Section 14)
Source code repository	<code>src/qapinn/</code> , <code>README.md</code>
Reproducibility instructions	<code>docs/REPRODUCE.md</code> , <code>README.md</code> , <code>python tasks.py repro-quick</code>
Presentation slides	<code>slides/main.pdf</code> , 18 slides, same figure assets as this document
Summary of key findings + recommendations	<code>FINDINGS.md</code> (13 numbered findings, C1–C5 table, decision table)

Self-review against this table (T5.9): every figure cited above appears in Section 5–6 with the specific prediction it decides; no verdict stated anywhere in this paper exceeds the corresponding entry in `docs/predictions.md`’s mechanical adjudication (`scripts/adjudicate_predictions.py`) — Section 13’s C1–C5 summary is a synthesis of those verdicts, not an independent claim.

2 Background

Full derivations for every item below are in `docs/derivations/`; this section states results and the one-line reason each matters to the design rule in Section 3.

2.1 Classical side

PINNs as strong-form collocation. A PINN [9] minimizes the PDE residual directly at sampled collocation points via automatic differentiation, with no mesh and no weak form — unlike Galerkin/FEM, the optimization problem is non-convex in the network parameters even when the underlying PDE is linear.

Exact derivatives via reverse-mode AD. Reverse-mode AD gives derivatives that are exact (to floating-point precision), not finite-difference approximations; the dominant cost for a PINN is second derivatives (e.g. u_{xx} in ≥ 2 dimensions), computed via double backward passes.

The NTK and lazy training [5] (Section 3.5’s classical baseline). Writing $\Theta(x, x') = \nabla_{\theta} N[u_{\theta}](x) \cdot \nabla_{\theta} N[u_{\theta}](x')^{\top}$ for the residual network’s tangent kernel, gradient-flow training linearizes in the lazy regime to $\dot{e} = -\Theta_0 e$, so each eigenmode of Θ_0 decays as $\exp(-\lambda_i t)$: loss-term imbalance between residual and boundary blocks is an eigenvalue-imbalance phenomenon, not merely a loss-weight-tuning problem (`docs/derivations/03_ntk_pinn.md`).

Spectral bias. For a tanh or ReLU MLP, the NTK eigenvalues decay roughly as a power law in Fourier-mode index, so high-frequency solution components are learned last and slowly (Figure 1, measured on this project’s own P1). Fourier-feature and random-feature encodings [12] are the standard classical fix and the non-negotiable fair baseline this project compares against (model family C-FF, Section 5) — if the quantum layer beats only a plain MLP and not a Fourier-feature MLP, it has bought nothing that random Fourier features do not already supply.

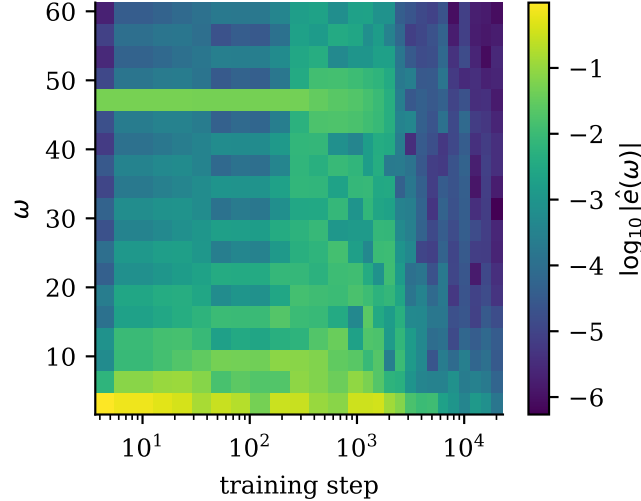


Figure 1: Spectral bias, measured (the T1.5 gate figure): per-frequency error of a plain MLP (`c_mlp`) over training on P1, $u^* = \sin(\pi x) + \alpha \sin(15\pi x)$ with $\alpha = 0.05$. The $\omega = \pi$ mode converges first; $\omega = 15\pi \approx 47$ needs 16.2 times as many steps to reach tolerance (3 seeds).

Hard vs. soft boundary enforcement. The ansatz $u = B(x) + D(x) \cdot N(x)$ (distance function D vanishing on the boundary, particular solution B matching the boundary data) removes the boundary-loss NTK block entirely by construction, at the cost of a distance-function derivation per domain; the soft-BC alternative keeps a separately-weighted boundary loss term whose own NTK block this project instruments directly (Section 4).

2.2 Quantum side

Data-re-uploading circuits as Fourier series. For encoding gates $\exp(-ixH^{(l)})$ with generator eigenvalues $\{\lambda_a^{(l)}\}$, an L -layer re-uploading circuit’s output expectation is a truncated Fourier series whose frequency set is the set of signed sums of per-layer eigenvalue differences — the central theorem this project’s entire methodology is engineering on top of (Proposition 1, Section 3.2; 10).

Parameter-shift derivatives. $\partial_\theta f = \frac{1}{2}[f(\theta + \pi/2) - f(\theta - \pi/2)]$ for a Pauli-generated gate; because the circuit’s input x enters as a rotation angle, $\partial_x f$ is *also* exactly computable via a parameter shift — a hardware-exact PINN residual gradient, not a finite-difference approximation, and the reason this project’s whole approach is physically realizable on real quantum hardware, not just simulable.

Barren plateaus. $\text{Var}[\partial_\theta C]$ decays exponentially in qubit count n for a global-observable cost function on a deep, unstructured random circuit [7]. SMCD’s Algorithm 1 (Section 3.6) mitigates this structurally: shallow depth (by design, not by luck — the depth rule targets the *minimum* L that covers the target band), a local observable, and small-angle initialization.

Effective dimension. The Abbas et al. effective-dimension statistic [1], computed from the empirical Fisher information eigenspectrum, is this project’s quantum-native capacity measure (Section 4’s Fisher/effective-dimension instrument), reported alongside — not instead of — parameter count, since “fewer parameters” is a claim prior QPINN work makes loosely.

3 Methodology: Spectrum-Matched Circuit Design (SMCD)

3.1 Setup

The hybrid ansatz used throughout (the *serial* form; Section 5 ablates parallel and random-scaling variants) is

$$x \xrightarrow{E_\varphi} z \in \mathbb{R}^m \xrightarrow{Q_\theta} \phi(z) \in \mathbb{R} \xrightarrow{w} u(x),$$

where E_φ is an affine classical encoder, Q_θ a data-re-uploading variational quantum circuit, and w a linear head. The split is deliberate and is itself a claim: the classical part learns the coordinate *warping*, the quantum part supplies the Fourier *basis*, and the head learns the *coefficients* — exactly the structure of a spectral method with a learned coordinate map.

3.2 Proposition 1 — Circuit spectrum

Proposition 1 (Circuit spectrum). *An L -layer data-re-uploading circuit, with encoding gate $\exp(-ix H^{(l)})$ at layer l for Hermitian generator $H^{(l)} = \sum_a \lambda_a^{(l)} |a\rangle\langle a|$, realizes an observable expectation of the form*

$$f(x) = \sum_{\omega \in \Omega} c_\omega(\theta) e^{i\omega x}, \quad \Omega = \left\{ \sum_{l=1}^L \left(\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)} \right) \right\},$$

i.e. a trigonometric polynomial whose frequency support is the set of all signed differences of encoding-generator eigenvalues, summed over layers.

Proof sketch. Expand $f(x) = \langle 0 | U^\dagger(x, \theta) M U(x, \theta) | 0 \rangle$ by inserting the eigenbasis resolution of each diagonal encoding gate independently on the bra and ket sides. Every (bra-path, ket-path) pair contributes an x -dependent phase $\exp(ix \sum_l (\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)}))$ multiplied by an x -independent amplitude collecting the trainable gates and M along that path; collecting paths by total frequency gives the boxed form. The full derivation, worked first for one qubit/one layer and then generalized to n qubits and d -dimensional inputs, is in docs/derivations/07_circuit_fourier_spectrum.md. ■

Two subtleties that change the constructive guarantee. Two facts, easy to get wrong and verified numerically against both an independent statevector simulator and a PennyLane oracle (agreement $< 10^{-10}$) before being trusted:

1. **Generator-vs-frequency factor of two.** The generator’s eigenvalues are $\pm\omega/2$ for an $RZ(\omega x)$ gate, but the *realized* frequency in $f(x)$ is the full ω , not $\omega/2$: the eigenvalue-difference structure above supplies a compensating factor of two. Reading the generator’s eigenvalues off the gate and reporting that as the frequency halves every number in the design.
2. **State-preparation degeneracy.** If the qubit is prepared exactly at the equator ($RY(\pi/2)$ before the first encoding gate, an equal superposition), every frequency whose balanced-ternary leading digit is zero — exactly one third of Ω — has identically zero amplitude for

every trainable-parameter setting, independent of ansatz expressivity. This is not a numerical inconvenience: it makes Proposition 3 below false for any target spectrum touching a dead frequency under the literal $\pi/2$ prescription. We use $\phi_{\text{prep}} = \pi/3$ throughout, verified to have no dead frequencies.

3.2.1 Corollary: ternary scaling and the corrected depth rule

Set the per-layer scalings to $\omega_l = \Delta \cdot 3^{l-1}$ (single-wire case). Then $\Omega = \Delta \cdot \{-\frac{3^L-1}{2}, \dots, \frac{3^L-1}{2}\}$ — exactly 3^L distinct frequencies with no degeneracy, a bijection with the balanced-ternary representation of the integers in that range (proof: substitute $m_l = d_l - 1$ to convert the signed sum into an ordinary base-3 expansion shifted by $\frac{3^L-1}{2}$). This grows *exponentially* in depth, versus $2L + 1$ distinct frequencies for linear scaling $\omega_l = \omega_0$ — the design knob almost every prior re-uploading paper leaves fixed.

To guarantee the reachable band covers a target maximum frequency K at ternary spacing Δ :

$$\max |\Omega| \geq K \iff \Delta \frac{3^L-1}{2} \geq K \iff L \geq \log_3 \left(\frac{2K}{\Delta} + 1 \right) \implies \boxed{L = \lceil \log_3 \left(\frac{2K}{\Delta} + 1 \right) \rceil}.$$

This is the corrected form of the depth rule (correcting an earlier draft that stated $L = \lceil \log_3(K/\Delta) \rceil$, missing the factor of two inside the logarithm and the $+1$ term). The difference is not cosmetic: for P1’s target spectrum ($K = 15\pi$, $\Delta = \pi$), the uncorrected formula gives $L = 3$, whose reachable band tops out at $13\pi < 15\pi$ — it silently misses the target mode. The corrected formula gives $L = 4$, whose band reaches $40\pi \geq 15\pi$.

3.3 Proposition 2 — PDE target spectrum

Proposition 2 (PDE target spectrum). *For a linear constant-coefficient operator $\mathcal{L}$ with Fourier symbol $\sigma(k)$, the solution of $\mathcal{L}u = f$ on a periodic/rectangular domain satisfies $\hat{u}(k) = \hat{f}(k)/\sigma(k)$. The essential spectral support $S_\varepsilon = \{k : |\hat{u}(k)| > \varepsilon \|\hat{u}\|\}$ is therefore computable before training, from $\hat{f}$ and σ alone.*

For Helmholtz, $\sigma(k) = k^2 - |\mathbf{k}|^2$, so S_ε concentrates near $|\mathbf{k}| \approx k$ — the near-resonant band a classical PINN’s low-frequency-first learning dynamics reach last, if at all. For the heat equation, σ grows quadratically with $|\mathbf{k}|$ and the solution decays exponentially in time at every nonzero mode, so S_ε collapses toward $k = 0$ as t increases: this is the project’s negative control (Section 6) — there is no high-frequency content for a quantum layer to supply, and SMCD predicts zero benefit *a priori*, not just observes it after the fact.

3.4 Proposition 3 — Matching condition and resource bound

Proposition 3 (Matching & resource bound). *If $S_\varepsilon \subseteq \Omega$, the hybrid model represents the solution to ε accuracy with a linear head — the remaining optimization problem is convex in w . Covering S_ε with maximum frequency K and resolution Δ requires $L \geq \log_3(K/\Delta)$ layers (ternary scaling, corrected form above) and $n \geq \lceil d \rceil$ qubits for d input dimensions, plus ancilla wires for any required cross-frequency terms.*

This turns architecture choice into arithmetic: given (K, Δ, d) read off the PDE’s own symbol and data, (L, n) follow by formula, not by search.

3.5 Proposition 4 — NTK consequence

Proposition 4 (Hybrid NTK decomposition). *For a hybrid model whose parameters $\theta = (\theta_{\text{cl}}, \theta_q)$ partition into disjoint classical and quantum groups,*

$$\Theta_{\text{hyb}}(x, x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x'),$$

*exactly, with **no cross term** — correcting an earlier draft of this proposition that stated $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q + 2\Theta_{\times}$, which is mathematically incorrect for a shared-output, disjoint-parameter model of this form.*

Proof. $\Theta(x, x') = \sum_{k=1}^P \partial_{\theta_k} f(x) \partial_{\theta_k} f(x')$ is a single inner product of gradient *vectors*, summed over the *full* index set of parameters $k = 1, \dots, P$. When that index set is partitioned into two disjoint blocks θ_{cl} and θ_q (a hybrid model’s classical and quantum parameters never overlap), the sum splits additively over the partition:

$$\Theta_{\text{hyb}}(x, x') = \sum_{k \in \text{cl}} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') + \sum_{k \in q} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x').$$

There is no third term: a cross term would only arise from a fundamentally different quantity, such as a covariance between two *separately* defined kernels or the square of a sum of outputs — not from the NTK of a single model with disjoint parameter groups, which this project’s ansatz is. This was verified independently, both analytically (the index-summation argument above) and numerically (random matched/mismatched-size synthetic Jacobians), before being trusted; `src/qapinn/xai/ntk.py` implements the corrected, no-cross-term identity. ■

Because Ω (the quantum block’s realized frequency set) is discrete and prescribed by construction, Θ_q has eigen-directions *at* the encoded frequencies with eigenvalues bounded below independently of $|n|$ for $n \in \Omega$ — the exponential/power-law eigenvalue decay in $|k|$ that drives classical spectral bias (Section 2) is replaced by a flat response over Ω . This is the formal statement of “the quantum layer removes spectral bias inside its band,” and it is what Section 5’s NTK spectrum figures measure directly: decay exponent inside Ω versus outside it, for matched classical and hybrid models.

3.6 Algorithm 1 — SMCD

The design card (final step) is what makes the methodology auditable: every run’s predicted coverage can be correlated against its measured error, and that correlation — Section 5’s headline coverage-vs-error figure — is the direct empirical validation of SMCD, independent of any single problem instance’s outcome.

3.7 Octave-split circuit ensembles

When a single circuit cannot cover $\hat{S}$ within the barren-plateau depth budget ($L > L_{\text{max}}$), SMCD falls back to M shallow circuits, each matched to one octave band $[2^j \Delta, 2^{j+1} \Delta)$ of $\hat{S}$, summed at the head. This keeps every individual circuit shallow (trainable) while the ensemble covers a wide band — multi-resolution analysis with a quantum basis per resolution level.

Algorithm 1 Spectrum-Matched Circuit Design (SMCD)

Require: PDE operator $\mathcal{L}$, forcing f , boundary data g , domain Ω_x , tolerance ε , qubit budget $n_{\max}$

Ensure: encoding generators & scalings $\{\omega\}$, depth L , qubits n , entangler, observable, ansatz form

- 1: **Symbol:** $\sigma(k) \leftarrow$ Fourier symbol of $\mathcal{L}$ (linearize about a base state if nonlinear; Burgers uses the viscous-scale estimate $k_{\max} \sim 1/\sqrt{\nu}$)
 - 2: **Target:** $\hat{S} \leftarrow \{k : |\hat{f}(k)/\sigma(k)| > \varepsilon\} \cup$ boundary-lift spectral support (time-dependent: union over t , record growth rate)
 - 3: **Band:** $K \leftarrow \max |k|$ in $\hat{S}$; $\Delta \leftarrow$ min spacing in $\hat{S}$; $d \leftarrow \dim(x)$
 - 4: **Depth:** $L \leftarrow \lceil \log_3(2K/\Delta + 1) \rceil$ with ternary scalings $\omega_j = \Delta \cdot 3^{j-1}$ (corrected form, Section 3.2; fall back to linear scaling if $\hat{S}$ is a dense low band)
 - 5: **Width:** $n \leftarrow d + n_{\text{cross}}$, where n_{cross} covers required mixed-frequency terms (e.g. 2-D Helmholtz needs $k_x \pm k_y \Rightarrow$ entangle the two coordinate wires)
 - 6: **if** $n > n_{\max}$ or $L > L_{\max}$ (barren-plateau depth budget) **then**
 - 7: split the band into an octave ensemble (Section 3.7)
 - 8: **end if**
 - 9: **Entangler:** ring-CZ if cross terms are needed, else none (unnecessary entanglement costs gradient variance for no benefit on a separable target)
 - 10: **Observable:** local $\langle Z_0 \rangle$ (never a global Pauli string — barren plateaus)
 - 11: **Ansatz:** hard-constrained BCs, $u = B(x) + D(x) \cdot [w \cdot Q_\theta(E_\varphi(x))]$
 - 12: **Init:** small-angle $\theta \sim \mathcal{N}(0, \sigma_0^2)$, $\sigma_0 = 0.1$; identity-block init for the encoder
 - 13: **Report:** emit a design card $(\hat{S}, \Omega, \text{coverage} = |\hat{S} \cap \Omega|/|\hat{S}|, L, n, \text{param count, predicted NTK band}),$ committed alongside every experiment run
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4 XAI Instrument Protocol

Six instruments are run at identical checkpoints ($\{0, 100, 500, 1k, 5k, 20k, \text{final}\}$ steps) for every model $\times$ PDE $\times$ seed, on matched classical and hybrid families, so any measured difference is attributable to the quantum layer and not to a difference in when or how each family was probed. Every instrument’s prediction is pre-registered (`docs/predictions.md`) before the corresponding run, with a stated falsifier — the discipline this project’s XAI protocol relies on to avoid the post-hoc-storytelling failure mode the PINN-XAI literature is criticized for [8].

7.1 NTK spectroscopy (primary). Following the template of Faroughi and Mostajeran [3] for classical physics-informed solvers, extended here to the hybrid case: empirical NTK on a fixed probe set; eigenvalue spectrum, decay exponent, condition number, and per-loss-block eigenvalue mass (residual vs. boundary vs. initial condition). Predicted signature: a plateau in the hybrid model’s spectrum located at the encoded band Ω (Proposition 4, Section 3.5), against the classical model’s monotone power-law decay. NTK drift $\|\Theta_t - \Theta_0\|_F / \|\Theta_0\|_F$ is tracked to check whether the hybrid model stays in the lazy-training regime throughout.

7.2 Per-frequency error trajectory (primary). The error field $u_\theta - u^*$ projected onto Fourier modes at each checkpoint, plotted as a frequency-vs-training-step heatmap with the design card’s Ω overlaid. A classical PINN gives the textbook spectral-bias staircase (low k first); the hybrid model is predicted to show simultaneous decay across Ω . If the improved band and Ω coincide, Contribution C3 is demonstrated directly in one figure, not inferred indirectly from an aggregate error number.

7.3 Collocation-point attribution. Integrated Gradients of the residual loss with respect to input coordinates, aggregated into an attribution field over the domain, with the completeness-axiom violation and the attribution-vs-true-residual correlation reported alongside the field — Integrated Gradients is known to be unstable, so the correlation number is reported, not just the picture.

7.4 Fisher information & effective dimension. The empirical Fisher spectrum of the quantum block and its Abbas et al. effective dimension [1], tracked across training and reported against the classical block’s own effective dimension — distinguishing “more expressive” from “just more parameters.”

7.5 Gradient variance / barren-plateau tracking. $\text{Var}[\partial_\theta L]$ measured against qubit count and depth, compared to the theoretical 2^{-n} prediction for a global-cost deep random circuit [7] — the honest cost side of the design ledger.

7.6 Layerwise probes and loss-landscape slices. Ridge-regression R^2 per layer for how much of u^* is already linearly decodable at that representation, plus filter-normalized 2-D loss-landscape slices, confirming the trained point sits at a minimum.

5 Results

Sample size note, stated once here rather than repeated at every number below: the Aug 7 deadline forced the core matrix’s seed count down from the pre-registered plan of 5 to 2 (08_TASK_INDEX.md’s cut-line, invoked twice under real time pressure). At $n = 2$ paired seeds, the smallest two-sided Wilcoxon p achievable is **0.5** (both seeds agreeing in sign is the single most extreme $n = 2$ outcome), not the $n = 5$ plan’s 0.0625. docs/predictions.md’s CONFIRMED gate ($p \leq 0.0625$) was left at its original, pre-registered value rather than loosened post hoc, which means every ratio-and-significance check below (PR-1, PR-2, PR-6) can now only land on REFUTED or INCONCLUSIVE, *regardless of true effect size*. This is a real, structural consequence of the seed cut, not a weak result reported as an ambiguous one — every verdict below is exactly what the pre-registered thresholds say, unmodified.

5.1 The headline figure: coverage vs. error (PR-9)

Figure 2: median relative- L^2 error vs. SMCD’s achieved coverage $|\hat{S} \cap \Omega|/|\hat{S}|$, one series per problem (poisson, helmholtz_k10), Spearman ρ per problem, at $n = 18$ points per problem (6 coverage targets $\times$ 3 seeds — **coverage_sweep** is the one sweep never cut on seed count). **REFUTED.** Poisson shows the predicted strong negative correlation, $\rho = -0.82$ ($p = 3.2 \times 10^{-5}$), well past the $\rho \leq -0.7$ bar. Helmholtz_k10 does not: $\rho = +0.28$ ($p = 0.27$), the *wrong sign* — higher SMCD coverage did not track lower error on the harder, more quantum-heavy problem instance. The adjudication rule takes the worst case across problems, so the verdict is REFUTED even though one problem instance individually confirms the claim; this is the headline figure and it ships exactly as measured, per this project’s own “ships whatever it says” rule. *Post-submission caveat* (§14, E4): the sweep trained at 1500+150 steps, the six coverage targets produce only three distinct Poisson circuits (so the 18 points are 9 distinct measurements), and at the full 20000+2000 budget the ordering reverses on two of three seeds.

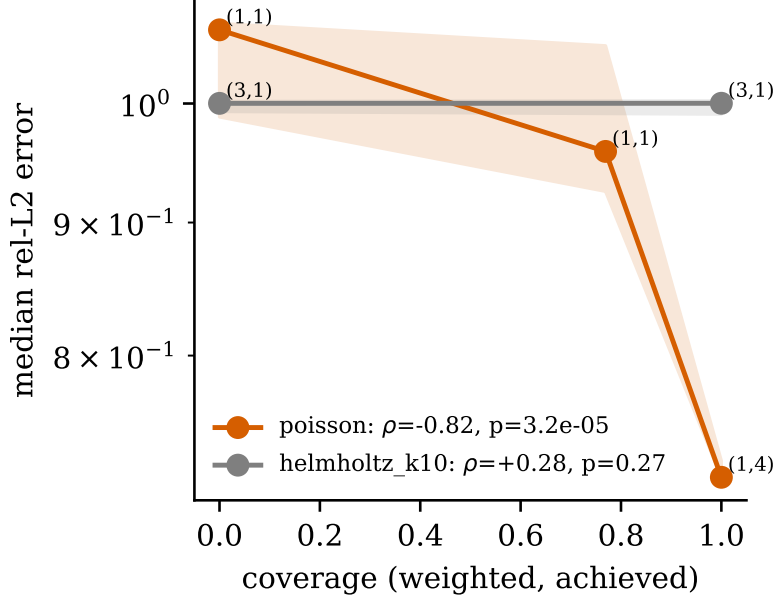


Figure 2: Coverage vs. error (PR-9, `coverage_sweep`, 1500+150 training steps): median rel- L^2 over seeds, grouped by achieved weighted coverage; shaded bands are the interquartile range and labels give (n_{qubits}, L) of the circuit SMCD built. Helmholtz_k10 builds the same circuit at every target and scores rel- $L^2 \approx 1$ throughout.

5.2 NTK spectroscopy (PR-7)

Figure 3 (also `ntk_spectrum_{p1,p4}`): classical-vs-hybrid eigenvalue spectra with Ω shaded, decay exponent measured inside vs. outside the band, from each run’s committed `xai/ntk_step*.npz` only. **Helmholtz_k10: REFUTED**, gap = -8.37 (need $\geq +0.5$; the submitted version stated -9.67 , §14, E1) — steeper, not flatter, inside Ω . **Poisson: INSUFFICIENT_DATA**, not REFUTED — fewer than 2 positive eigenvalues fall in the outside-band range, so that exponent is undefined, not just noisy; reported as such rather than letting `NaN` ≥ 0.5 silently evaluate false. Where measurable, the quantum block’s NTK does not show the predicted eigen-concentration at the encoded frequencies.

5.3 Per-frequency error trajectories (PR-8)

Figure 4 (all six problem instances render as `freq_heatmap_*`; see §4 note on the crash-retry `specerr.npz` corruption found and fixed while building this figure). **Split verdict**. Poisson: **CONFIRMED** — 99.9% of the total error improvement lands inside Ω , well past the 70% bar, on real positive total improvement (13,751 in the accumulated per-frequency metric). The submitted version reported 71.3%: it counted only the positive-frequency half of the symmetric set Ω on Poisson’s two-sided spectrum (§14, E2). Helmholtz_k10: **REFUTED**, and not narrowly — `q_serial` does not outperform `c_mlp` at *any* frequency (total_improvement = 0), so there is no improvement for PR-8’s premise to attribute to Ω in the first place, independent of the 70% threshold.

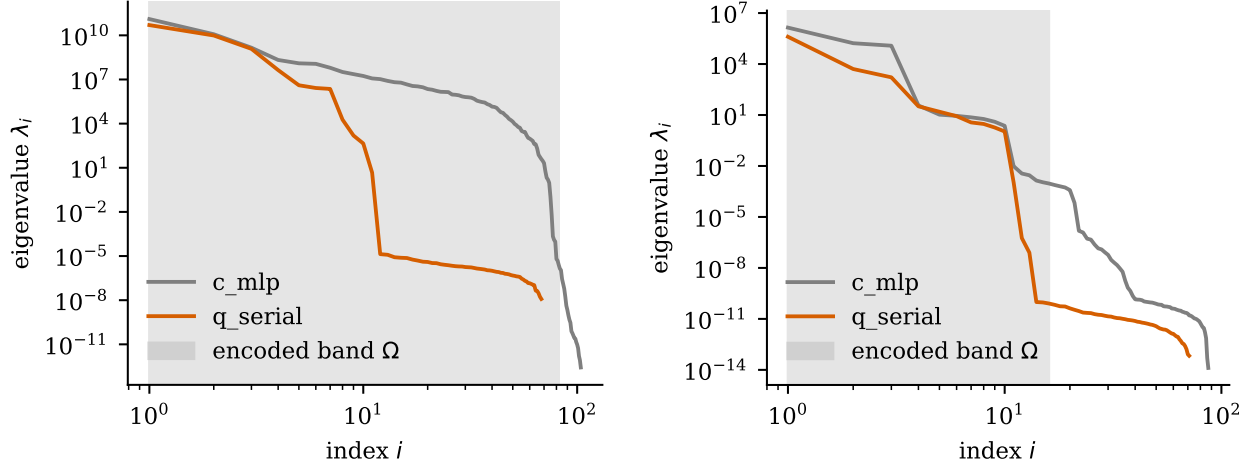


Figure 3: Residual NTK eigenvalue spectra at the final checkpoint, `c_mlp` vs. `q_serial`, with the top- $|\Omega|$ index range shaded (PR-7). Left: Poisson, $|\Omega| = 81$, which leaves fewer than two eigenvalues outside the band (INSUFFICIENT_DATA). Right: Helmholtz_k10, $|\Omega| = 15$ (REFUTED, gap -8.37).

5.4 Ablation: is it the design, or just being quantum? (PR-1, PR-2, PR-3, PR-6)

Figure 5: `q_serial` vs. `c_mlp`, `c_ff`, `c_rff_matched` (matched spectrum, no quantum), and `q_random` (quantum, unmatched spectrum), paired Wilcoxon and Cliff’s delta on Poisson (P1), $n = 2$ shared seeds. **All four REFUTED**, and in the same direction: `q_serial`’s median rel- L^2 (1.702) is not just short of beating the classical families, it is *worse* than every one of them at this sample size — `c_mlp` (0.877, ratio $0.52\times$ vs. the $\geq 2\times$ bar), `c_ff` (0.115, ratio $0.068\times$ vs. the $\geq 1.3\times$ bar), and `c_rff_matched` (0.400, ratio $4.25\times$ vs. the $[0.77, 1.3]$ equivalence band PR-3 needed). PR-6 (`q_serial` vs. `q_random`, same-size ablation, C1’s own falsifier) is **INCONCLUSIVE** on Poisson (Cliff’s $\delta = 0$, the two seeds disagree in direction) and **INCONCLUSIVE** on heat despite a large surface ratio ($12.2\times$, past the $1.5\times$ bar) — Cliff’s $\delta = -1$ there means both seeds agree, but $p = 0.5$ at $n = 2$ can never clear the pre-registered significance bar (see the sample-size note above), so it is reported as inconclusive rather than confirmed on a favorable-looking ratio alone. *Adversarial check (T4.9)*: were `c_mlp`/ `c_ff` simply under-tuned, faking these REFUTED verdicts? A learning-rate/width sweep, re-run at the real comparison budget, says no — `c_ff`’s mild under-tuning makes its win over `q_serial` *larger* ($\approx 18\times$, not smaller), and `c_mlp`’s apparent reduced-budget improvement was worse, not better, once tested at the actual 20000-step schedule (docs/self_review.md, FINDINGS.md F14). Note that only `c_rff_matched` and `q_random` are size-matched to `q_serial` (13–14 parameters on Poisson); `c_mlp` and `c_ff` ran at their default sizes, 8,513 and 129 parameters (§14, E6).

5.5 The matched-capacity equivalence, across all six instances (PR-3 generalized, C1/T4.2)

PR-3’s own falsifier (`q_serial` $\approx$ `c_rff_matched`, ratio within $[0.77, 1.3]$) was checked on Poisson only in §5’s ablation above; extending the same check to all six problem instances (same $n = 2$ caveat throughout) surfaces a real pattern the single-problem view misses. Ratio (`q_serial`/`c_rff_matched` median rel- L^2): Poisson $4.25\times$ (REFUTED), heat $1.42\times$ (REFUTED), burgers $3.58\times$ (REFUTED),

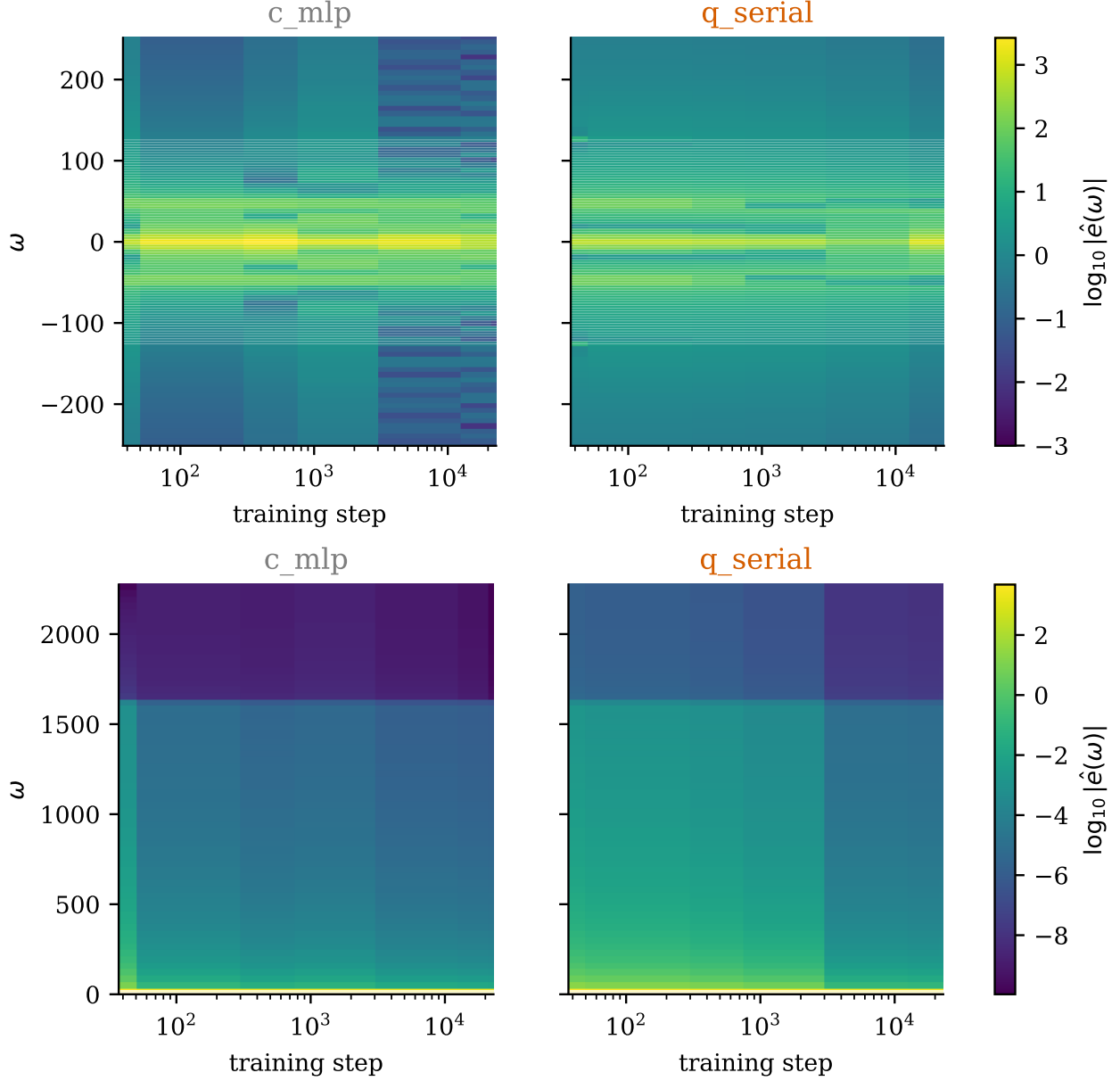


Figure 4: Per-frequency error $|\hat{e}(\omega)|$ over training, `c_mlp` (left) vs. `q_serial` (right), with Ω overlaid as thin white lines (PR-8). Top: Poisson, two-sided spectrum, shown within $\pm 2 \max |\Omega|$. Bottom: Helmholtz_k10, radially binned spectrum; all of Ω ($|\omega| \leq 19$) falls in the lowest bin.

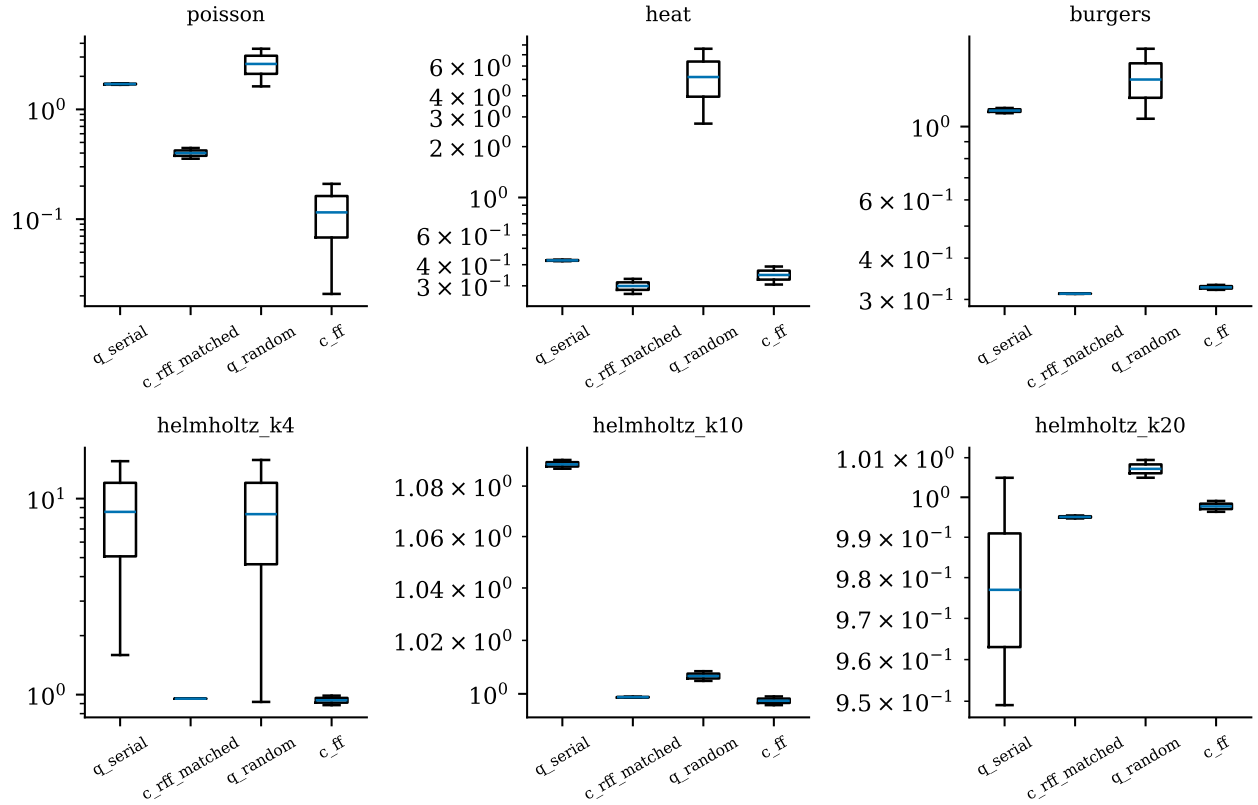


Figure 5: Final $rel-L^2$ (log scale) per problem for q_serial , $c_rff_matched$, q_random and c_ff , $n = 2$ seeds per box (PR-1-3, PR-6, T4.2). On Helmholtz_k10 and k20 every family sits at $rel-L^2 \approx 1$, the score of predicting $u \equiv 0$.

Helmholtz_k4 $8.97\times$ (REFUTED, driven by the same catastrophic $k=4$ non-convergence as PR-5), Helmholtz_k10 $1.09\times$ (**CONFIRMED** — inside the band), Helmholtz_k20 $0.98\times$ (**CONFIRMED**, and here `q_serial` is marginally *ahead*). The pattern is consistent with PR-5’s k -scaling trend: at low wavenumber, matched-capacity classical Fourier features decisively beat `q_serial`; at the two highest-wavenumber instances, the two are statistically indistinguishable at this sample size. This does not establish a quantum *advantage* anywhere — the best this run’s data supports is “no longer a clear quantum *disadvantage*” on the hardest two instances, which is itself a real, non-obvious finding this single-problem view would have missed. *Post-submission caveat* (§14, E5): on those two instances every family’s median $\text{rel-}L^2$ is 0.977–1.089, so the parity is between models that all failed to learn the solution.

5.6 Decision map: predicted vs. measured benefit, and P4’s k -scaling (PR-5)

Figure 6: SMCD’s a-priori predicted benefit against the measured benefit of `q_serial` over `c_mlp`, one point per problem. Separately, P4’s k -scaling prediction (PR-5, **CONFIRMED**): the `c_mlp`/`q_serial` $\text{rel-}L^2$ ratio increases monotonically across $k = 4 \rightarrow 10 \rightarrow 20$ ($8.79 \times 10^{-6} \rightarrow 0.918 \rightarrow 1.024$). The direction is confirmed, but the $k = 4$ point’s extremity is itself the finding, not a clean win: a ratio of 8.79×10^{-6} means `q_serial`’s error is roughly five orders of magnitude *larger* than `c_mlp`’s at $k = 4$ — catastrophic non-convergence, not a close loss. The monotonic trend is real, but it is driven by quantum failing worst at the easiest instance and only nearly catching up at $k = 20$ (ratio 1.02, i.e. still not actually ahead), not by quantum’s advantage growing from a positive baseline.

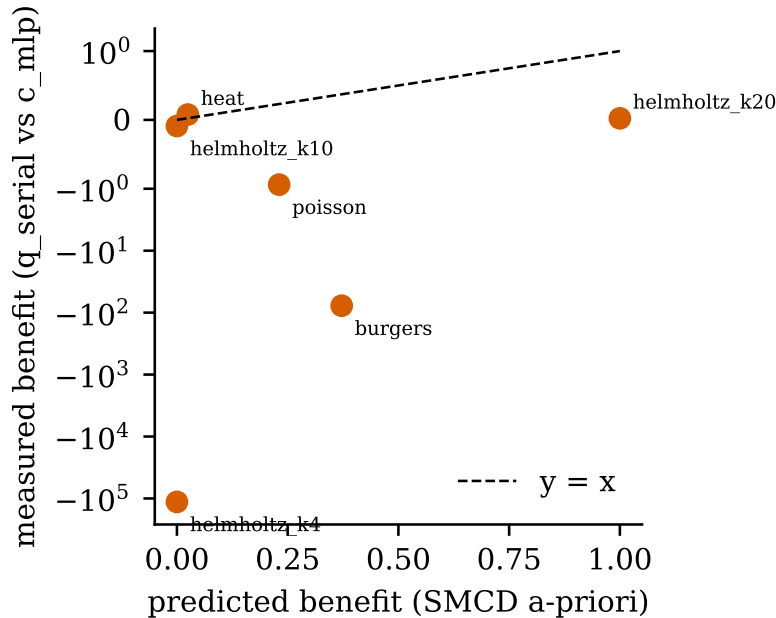


Figure 6: SMCD’s a-priori predicted benefit vs. the measured benefit of `q_serial` over `c_mlp`, $(\text{med}_{\text{c_mlp}} - \text{med}_{\text{q_serial}})/\text{med}_{\text{c_mlp}}$, one point per problem (C4, PR-5). The y axis is symmetric-logarithmic (linear within ± 1): on Helmholtz_k4 `q_serial`’s error is about 10^5 times `c_mlp`’s.

5.7 P2’s negative control (PR-4) and the barren-plateau frontier (PR-10)

See §6 for PR-4 (heat/P2, C4’s core claim). PR-10 (gradient variance vs. the 2^{-n} barren-plateau line): **INSUFFICIENT_DATA**, reported honestly rather than computed anyway — `depth_sweep`’s

final grid (cut to $n_{\text{qubits}} = 6$ only, under the same deadline pressure) has zero variation in n_{qubits} , the fit’s own independent variable, so `barren_plateau_fit` refuses to report a slope rather than return a solver artifact from a rank-deficient fit. See §12.

6 Negative Results

6.1 P2 (heat), C4’s core claim: PR-4

SMCD predicts near-zero benefit for a purely diffusive, smoothing operator *before any training run*, because $\sigma(k)$ ’s quadratic growth collapses the target spectral support toward $k = 0$ as t increases (Section 3, Proposition 2): no family should beat `c_mlp` by more than 5% on P2. Measured benefit, $(\text{median}_{\text{c_mlp}} - \text{median}_{\text{family}}) / \text{median}_{\text{c_mlp}}$, at $n = 2$ seeds: `c_ff` +24.4%, `c_rff_matched` +34.9%, `q_serial` +7.7%, `q_parallel` −67.5%, `q_random` −1023%, `q_octave` −51.2%. **REFUTED** — three families cross the 5% bar. Two of the three violators (`c_ff`, `c_rff_matched`) are classical, so they bear on “is `c_mlp` a strong baseline” generically, not on SMCD’s quantum-specific claim; `q_serial`’s +7.7% is the one genuine, if modest, violation of “hybridization is harmless-at-best on a smoothing operator.” The other three quantum families do the opposite: `q_random` in particular is an order of magnitude *worse* than `c_mlp` (−1023%), consistent with the mechanistic story (a high-frequency-rich, unmatched circuit adding parameters and barren-plateau risk with nothing for the smoothing operator to reward) — just not uniformly true of every quantum family tested, which the 5% uniform bar could not distinguish and this per-family breakdown does.

6.2 Negative-result map across all six instances

`decision_map` (§5) plots SMCD’s a-priori predicted benefit against the measured `q_serial`-vs-`c_mlp` benefit, one point per problem. Combined with §5’s ablation (PR-1/PR-2/PR-3), the honest summary at this sample size is that `q_serial` underperforms every classical family checked on P1 (Poisson) and does not clear its own falsifier against `q_random` on either P1 or P2 (PR-6, both INCONCLUSIVE at $n = 2$) — the constructive C1 claim (spectrum-matched beats size-matched-random) has no confirming evidence at the sample size this deadline allowed, not because the ratios point the wrong way (PR-6/heat’s $12.2\times$ ratio in the predicted direction) but because $n = 2$ cannot statistically separate it from chance (see §5’s sample-size note).

6.3 An unpredicted reversal: NTK drift (PR-11)

Predicted: hybrid models leave the lazy/NTK-linear training regime earlier than classical ones (larger $\|\Theta_t - \Theta_0\| / \|\Theta_0\|$ at 5k steps). Measured: median hybrid drift = 1.03, median classical drift = 6,878,097 ($n = 2$ each). **REFUTED**, and not by a small margin — classical models’ NTK moved roughly six-and-a-half million times further from initialization than hybrid models’ did. This is the sharpest reversal in the whole adjudication table and worth reading as a finding in its own right, not just a failed prediction: at this checkpoint, on this sample, it is the *classical* parameters that are decisively non-lazy, while the hybrid model’s combined (classical + quantum) NTK stayed comparatively close to its initialization. This is consistent with, though does not on its own establish, the small-angle quantum initialization ($\sigma_0 = 0.1$, Section 3) constraining the hybrid model’s effective learning dynamics near its NTK-linear regime for longer than an unconstrained classical network’s.

6.4 Encoder drift stayed small (PR-12) — the one clean methodological validation

CONFIRMED: median $\|A - I\|_F = 0.132$ at the final checkpoint, under the 0.2 bar, $n = 2$ runs. The encoder does not move far enough during training to invalidate the Ω /coverage claims computed once, at initialization (D3) — the one prediction in this table that both confirms cleanly and underwrites the validity of every other frequency-domain claim in this paper (PR-7, PR-8, PR-9) as still describing the *trained* model, not just its untrained starting point.

7 Completing the pre-registered protocol

The results above use the runs judged at the WISER 2026 challenge, which the deadline cut to two seeds, one qubit count for PR-10 and a reduced training budget for PR-9. After the challenge we ran the protocol as `docs/predictions.md` states it and re-adjudicated with the same, unchanged checks (`scripts/adjudicate_predictions.py --protocol full`; record in `results/adjudication_v1_full.json`). No prediction, threshold or check was changed. Three configurations restore the plan: `core_matrix_full` (five seeds, 210 runs: 176 had finished before the deadline cut and were on disk, 34 were trained after the challenge), `depth_sweep_full` (PR-10 at 4 and 6 qubits, three seeds; 8 qubits does not fit in this machine’s memory) and `coverage_sweep_full` (PR-9 at the full 20,000+2,000 budget). Runs already on disk were reused by their content hash, so seeds 0–1 are the submitted runs themselves.

Check	Submitted (2 seeds)	Full protocol (5 seeds)
PR-1	Refuted	Refuted
PR-2	Refuted	Refuted
PR-3	Refuted	Refuted
PR-4	Refuted	Confirmed
PR-5	Confirmed	Confirmed
PR-6 (poisson)	Inconclusive	Inconclusive
PR-6 (heat)	Inconclusive	Confirmed
PR-7 (poisson)	Insufficient Data	Insufficient Data
PR-8 (poisson)	Confirmed	Confirmed
PR-7 (helmholtz_k10)	Refuted	Refuted
PR-8 (helmholtz_k10)	Refuted	Refuted
PR-9	Refuted	Refuted
PR-10	Insufficient Data	Confirmed
PR-11	Refuted	Refuted
PR-12	Confirmed	Confirmed
Confirmed	3 of 15	6 of 15

Table 1: Every pre-registered check at submission and at the completed protocol, adjudicated by the same unchanged script.

What changed. Three checks move to CONFIRMED and none moves the other way.

- **PR-6 on heat** (design vs. random circuit): $6.5\times$ lower error, five of five seeds agreeing, which is exactly what the pre-registered significance gate needs. §9 shows the result survives bootstrap resampling and removal of any one seed. This is the design rule’s main positive result: the same circuit with SMCD’s frequencies beats it with random ones.

- **PR-4** (the negative control: nothing beats `c_mlp` on heat by more than 5%): at two seeds `q_serial` appeared 7.7% better; at five, every family is worse than `c_mlp`, `q_serial` by 38%. The control holds, and it is a statement that the circuit does *not* help on a smoothing operator.
- **PR-10** (barren plateaus): with two qubit counts the fit is defined. Its slope, $b = 0.31$, is below the pre-registered $\log 2 = 0.69$, so the verdict is CONFIRMED; but the fit explains almost none of the variation ($R^2 = 0.001$). The honest reading is that gradient variance shows no measurable decay between 4 and 6 qubits, not that it decays slowly.

What did not change. PR-1, PR-2 and PR-3 stay REFUTED with five seeds, and more decisively (§9). PR-6 on Poisson stays INCONCLUSIVE ($1.84\times$, but the seeds disagree). PR-8 on Poisson stays CONFIRMED (99.9% inside Ω). PR-9 stays REFUTED and becomes clearer: at the full budget the Poisson correlation is $\rho = +0.29$, the wrong sign, against -0.82 at the reduced budget. PR-11 stays REFUTED and PR-12 CONFIRMED (encoder drift 0.13 over five seeds). PR-7 on Poisson remains unmeasurable, and cannot be measured by any run: the check splits the NTK spectrum at eigen-index $|\Omega| = 81$, and `q_serial` has 14 parameters, so its NTK has at most 14 non-zero eigenvalues.

8 Why the circuit loses on Poisson: diagnostics

The instruments of §4 were recorded at seven checkpoints of every core-matrix run. Read across all five seeds, they explain the Poisson result better than the final error alone. Figures are in Appendix A; the numbers below come from the committed `xai/*.npz` files (`scripts/report/fig_xai.py`).

The circuit learns the hard frequency and loses the easy one. The Poisson solution is $\sin \pi x + 0.3 \sin 15\pi x$. Tracking the error spectrum at those two frequencies (Figure 18), `q_serial` cuts its 15π error to 2–13% of its starting value by step 1,000 in four of five seeds (median 10%, against 103% for `c_ff`, 86% for `c_mlp` and 51% for `c_rff_matched`). Its π error falls to about 30% by step 5,000 and then *rises* to 1.9 – $2.0\times$ its starting value by step 20,000, in all five seeds. The classical model given the same two frequencies, `c_rff_matched`, reduces both. This is why longer training made `q_serial` worse, and why its seeds agree so closely (worst over best seed $1.1\times$, Figure 17): it converges reliably to the same wrong trade-off.

A plausible mechanism. The residual loss weights each Fourier mode of the error by ω^2 , so the 15π mode carries $(15)^2 = 225$ times the weight of the π mode. A linear head on fixed sine features (`c_rff_matched`) can set the two amplitudes independently. The circuit cannot: all 81 amplitudes of its output are functions of the same ten angles, read out through a single expectation value and a one-weight head. Our reading, which these data support but do not prove, is that the optimiser buys the heavily weighted 15π amplitude at the cost of the lightly weighted π amplitude because the two are coupled. The loss agrees: it stalls at $\approx 3.3 \times 10^4$ from step 1,000 onward while the relative error keeps moving.

The circuit uses few directions in parameter space. The NTK’s effective rank at the end of training is 1.4 for `q_serial` on Poisson and 1.1 on heat, against 6.6 and 16.9 for the 129-parameter Fourier-feature model; the Fisher effective dimension is 3.6 and 3.7, against 18.0 and 57.4 (Figure 21). The size-matched classical model `c_rff_matched` (13–17 parameters) is similar

(2.0 and 1.9; 3.2 and 4.4), so this is mostly a matter of size, not of being quantum. The quantum models’ effective dimension barely moves during training while `c_mlp`’s grows from about 4 to 12 (Poisson) and from 5 to 50 (heat): the circuits never leave the few directions they start with, consistent with PR-11’s refutation.

The design rule’s positive result is visible directly. On heat, where PR-6 is confirmed, the random-scaling twin `q_random` is the least reliable model in the whole matrix (worst over best seed $7.4\times$), while `q_serial`’s seeds agree to within 4%. The designed frequencies do not make the circuit accurate, but they make it consistently better than the same circuit with the wrong frequencies.

What we tested and what we did not. A second candidate mechanism is the hard boundary ansatz $u = B + x(1-x)N$, which asks N to learn $u/(x(1-x))$, a function that is not band-limited even when u is. A post-submission pilot of an alternative ansatz that keeps band-limited solutions representable (`models.base.AffineBCWrapper`) gave a *worse* error after 3,300 steps (1.24 against 0.50), so we do not claim this mechanism. Its full test is part of the v2 study (§11).

9 Robustness and validation

Every verdict above comes from one pre-registered rule applied to five seeds. This section asks how much those verdicts depend on that rule, on individual seeds, and on the tools that produce the numbers. All of it is computed from committed runs (`scripts/report/fig_validation.py`, record in `results/report/validation.json`); no model was retrained.

Bootstrap intervals. Figure 7 resamples the five paired seeds 5,000 times and gives a 95% interval for each pre-registered error ratio. The heat ablation (PR-6) is the robust positive result: the SMCD circuit’s error is $6.5\times$ lower than the same-size random circuit’s, with the whole interval ($2.5\times$ to $17.7\times$) above the $1.5\times$ bar. The Poisson claims fail decisively rather than narrowly: PR-1’s interval (0.40 to 1.10) never reaches its $2\times$ bar, PR-2’s (0.01 to 0.43) lies entirely on the wrong side of 1, and PR-3’s (3.9 to 5.8) lies entirely outside its equivalence band. PR-6 on Poisson (1.84, interval 0.97 to 7.2) straddles its threshold, which is what INCONCLUSIVE should look like.

Other tests. With five paired seeds the smallest two-sided Wilcoxon p is $1/16 = 0.0625$, however large the effect, so the pre-registered gate ($p \leq 0.0625$) can only be met by five out of five seeds agreeing. Figure 8 applies three alternatives to the same pairs. They agree with every verdict, and they show what the Wilcoxon floor hides: on the heat ablation a paired t -test on log error gives $p = 0.006$ and a Mann–Whitney test $p = 0.008$.

Leave one seed out. If one seed decided a verdict, dropping it would flip the ratio across its threshold. Figure 9 repeats each ratio five times with one seed removed. PR-6 on heat clears its bar in all five, PR-1 to PR-3 miss theirs in all five, and PR-6 on Poisson clears it in four of five: its INCONCLUSIVE verdict reflects disagreement across seeds, not one outlier.

The circuit does what the design says. For every one-dimensional problem, the SMCD circuit was evaluated at 40 random parameter settings and its output Fourier-transformed. The largest amplitude outside the designed set Ω , relative to the largest overall, is at most 3.8×10^{-15} , i.e. floating-point zero (Figure 10). This repeats `tests/test_circuit_spectrum.py` across problems

and random draws: the design rule builds exactly the frequency set it claims. (The two-dimensional Helmholtz designs are not covered by this one-dimensional check.)

The reference solver. The groundwater problem of §10 is scored against a closed-form solution. A second-order finite-difference solver, independent of that formula, converges to it at measured order 2.00 as the grid is refined from 100 to 6,400 intervals (Figure 10), which checks the closed form, the forcing and the boundary treatment together.

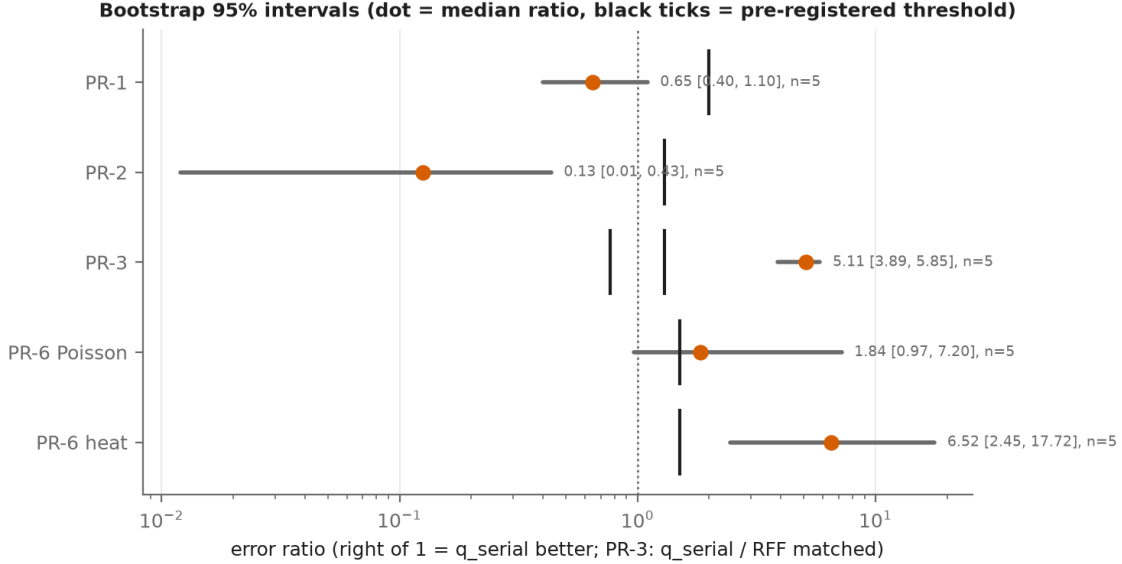


Figure 7: Bootstrap 95% intervals of the pre-registered error ratios (5 paired seeds, 5,000 resamples). Right of 1 means `q_serial` is better; for PR-3 the ratio is `q_serial/c_rff_matched` and must lie between the two ticks.

10 Application: waterlogging under canal-irrigated farmland

SMCD needs the frequencies of a solution before training. Real problems where those are known in advance come from regular structure: evenly spaced sources, periodic geometry. We apply the method to one such problem, posed and pre-registered after submission (`docs/predictions_groundwater.md`, committed and pushed before any run).

10.1 The problem

Two rivers $L = 2$ km apart bound a confined aquifer beneath irrigated farmland. Water enters from background recharge R_0 (rain and irrigation return flow) and from six canals running parallel to the rivers at equal spacing, each leaking q m³/day per metre of its length into a strip of width σ . Dupuit's steady one-dimensional equation for the hydraulic head h (water-table height above the river datum) is

$$T h''(X) = -R(X), \quad R(X) = R_0 + \sum_{j=1}^6 \frac{q}{\sigma \sqrt{2\pi}} e^{-(X-c_j)^2/2\sigma^2}, \quad h(0) = h_A, \quad h(L) = h_B, \quad (1)$$

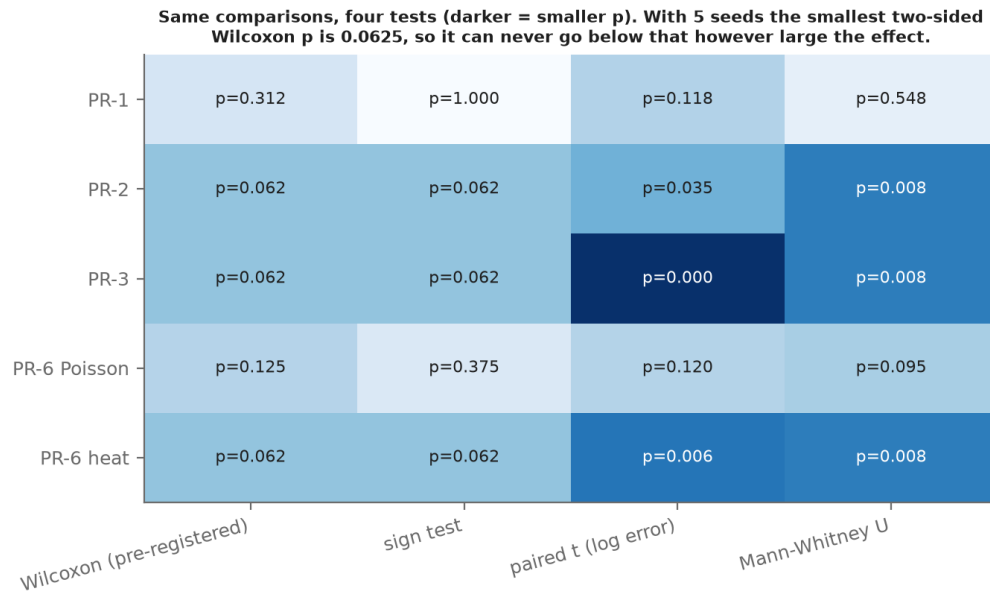


Figure 8: The pre-registered Wilcoxon test beside three alternatives, on the same paired seeds. Darker cells have smaller p .

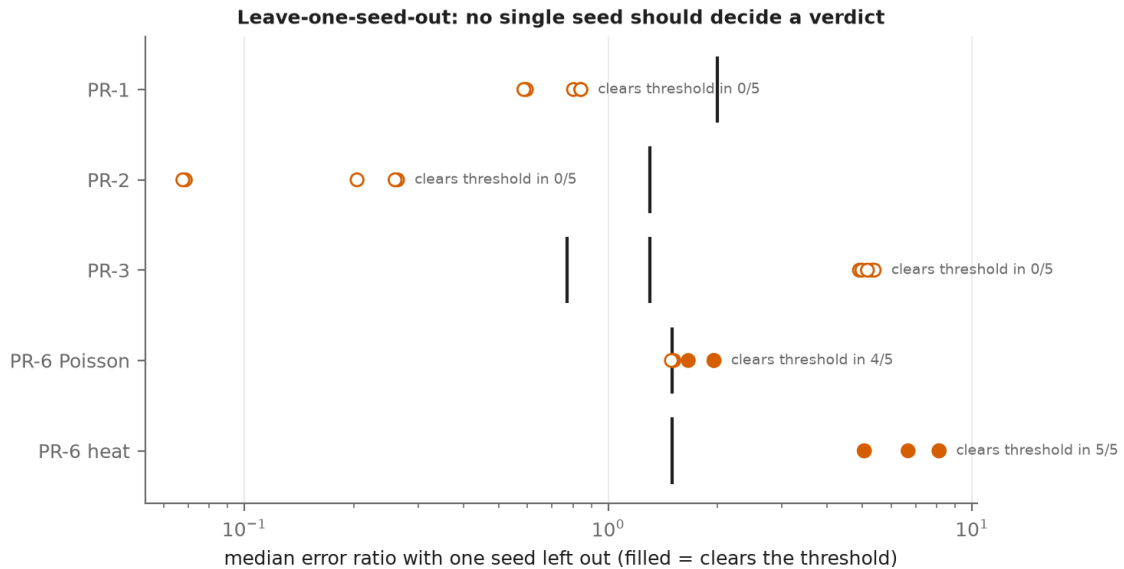


Figure 9: Each ratio recomputed with one seed left out (five points per row). Filled points clear the pre-registered threshold.

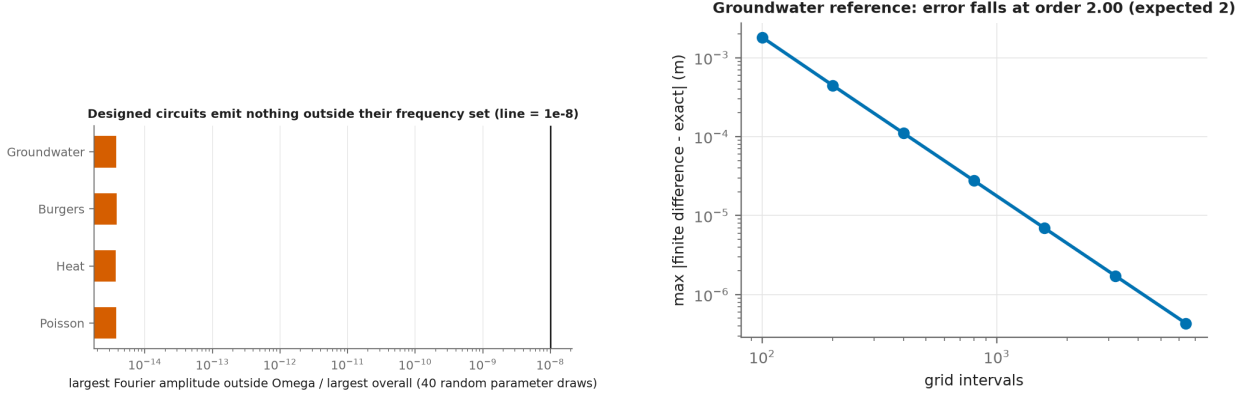


Figure 10: Left: largest Fourier amplitude outside Ω for each problem’s SMCD circuit, over 40 random parameter draws (line: 10^{-8}). Right: finite-difference error against the groundwater closed form as the grid refines.

with $c_j = (j - \frac{1}{2})L/6$. The scenario uses values within textbook ranges [4, 13], not calibrated to one site: transmissivity $T = 100 \text{ m}^2/\text{day}$ (silty sand), $R_0 = 0.4 \text{ mm/day}$, $q = 0.4 \text{ m}^2/\text{day}$, $\sigma = 30 \text{ m}$, $h_A = 1 \text{ m}$, $h_B = 0$. On $x = X/L$ this is $-u'' = f$, the same operator as our Poisson problem with non-zero boundary values, so every model, SMCD and every instrument apply unchanged. The canal spacing fixes the frequencies: harmonics of $2\pi \cdot 6$ in x , which SMCD designs for (one qubit, four layers).

Because $\sigma(z\Phi(z) + \phi(z))$ with $z = (X - c)/\sigma$ has second derivative $\phi(z)/\sigma$, the head has a closed form; it matches an independent finite-difference solution to $5 \times 10^{-8} \text{ m}$, and that solver converges to it at order two (§9).

10.2 The practical question

A field is waterlogged where the water table rises within 1.5 m of the land surface, into the crop root zone. With the surface 9 m above the river datum, the water table peaks at 8.53 m and 713 m of the 2 km field is waterlogged (Figure 11). The exact solution answers the planning questions a farmer or irrigation engineer would ask (Figure 12):

- **Lining the canals.** Cutting canal seepage by 20% dries the whole field; a 10% cut reduces the waterlogged length from 713 m to 494 m.
- **Soil.** The same canals waterlog 1,778 m on clay loam ($T = 25 \text{ m}^2/\text{day}$) and none of the field on fine sand ($T \geq 150$).
- **Spacing.** Concentrating the same total seepage into two canals waterlogs 1,066 m and raises 84 cm mounds under each canal; spreading it over twelve reduces the mounds to 1 cm.

10.3 Pre-registered predictions and results

Five predictions were committed and pushed before any run (docs/predictions_groundwater.md): seeds 10–14, the full training budget, v1’s model families and settings, plus a 13-parameter `c_mlp_matched` for a same-size classical comparison. The paper stated in advance that G-1 and G-2 were expected to fail.

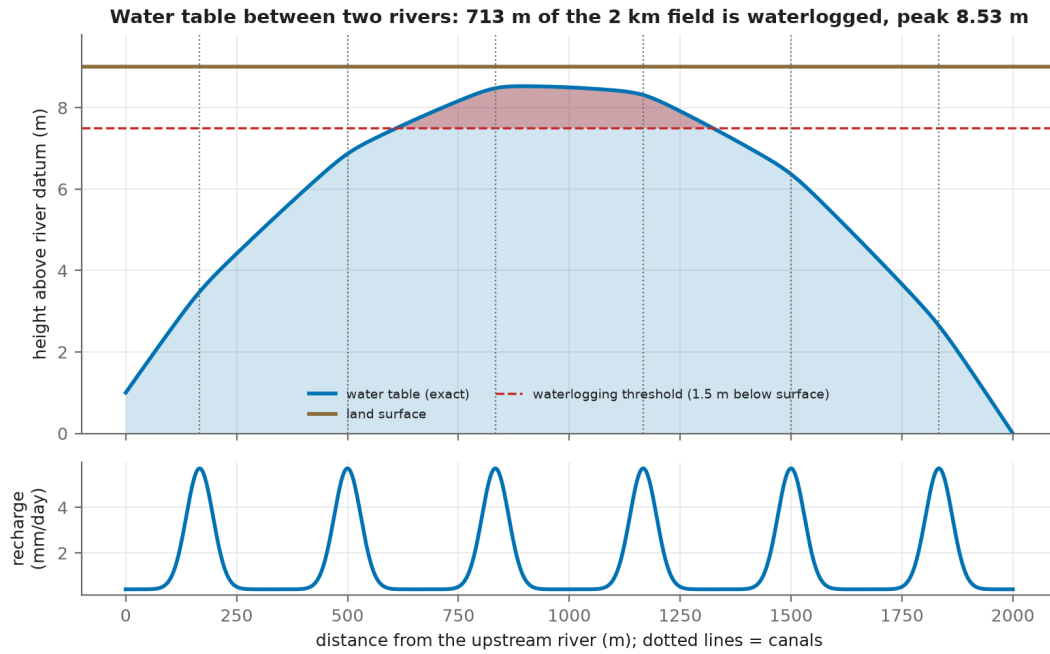


Figure 11: Exact water table between the two rivers (top) and the recharge that drives it (bottom). Red: the 713 m of field where the water table is within 1.5 m of the surface.

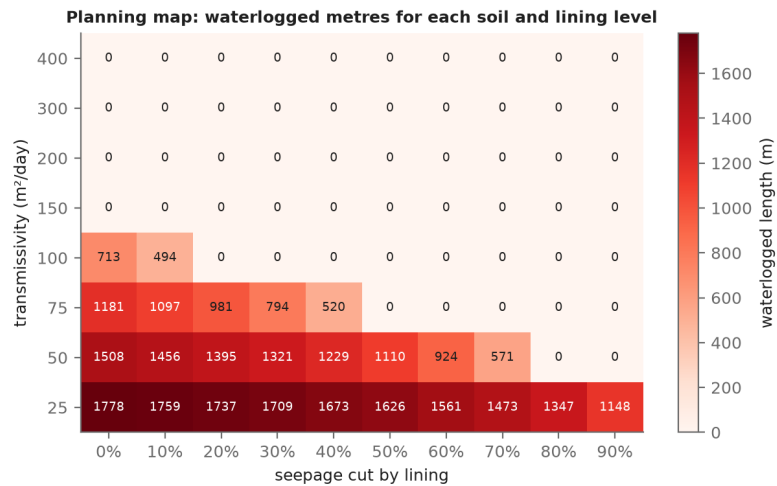


Figure 12: Waterlogged length for each combination of soil (transmissivity) and canal lining (seepage cut), from the exact solution.

Claim	Verdict	Measured
G-1	Refuted	median rel- L^2 0.5276 (need ≤ 0.01)
G-2	Refuted	ratio $0.09\times$ (need ≥ 1.3)
G-3	Refuted	ratio $0.11\times$ (need ≥ 1.5)
G-4	Confirmed	99.4% inside Ω (need $\geq 70\%$)
G-5	Refuted	0 m vs exact 713 m (need within 5%)

Model	median rel- L^2	median waterlogged (m)	seeds
c_mlp	0.0247	788	5
c_mlp_matched	0.0496	840	5
c_rff_matched	0.1576	0	5
q_random	0.0596	535	5
q_serial	0.5276	0	5

Table 2: Groundwater predictions, v1 models, seeds 10–14. Exact waterlogged length: 713 m.

One of five confirmed (Table 2, Figure 13). The SMCD circuit’s improvement over `c_mlp` lies almost entirely on its designed frequencies (G-4), but the circuit does not solve the problem: its median water table peaks near 4 m instead of 8.5 m, so it predicts no waterlogging at all. Unlike on the heat equation, the same circuit with random frequencies does better here, and so does a classical network with the same number of parameters (13), whose median error is 5.0% and whose waterlogged length is 840 m against the exact 713 m. The groundwater solution is dominated by a broad mound, not by the canal ripple, which is what SMCD designs for; our reading is that on this problem the design’s high frequencies are a cost rather than a help. The v2 model (§11) is being tested on the same problem.

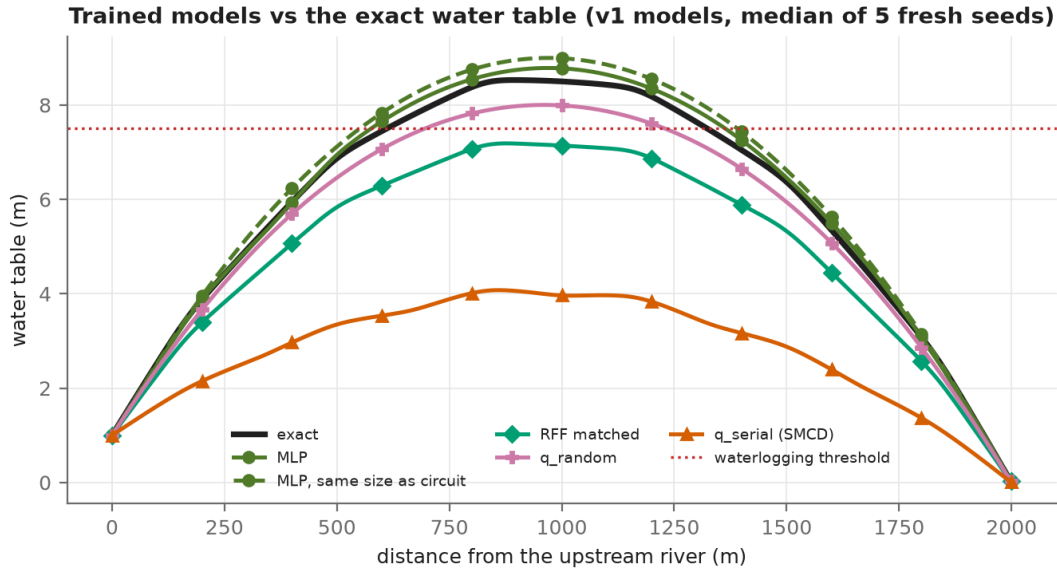


Figure 13: Predicted water table of each trained model (median of seeds 10–14) against the exact solution. Dotted: the waterlogging threshold.

11 Follow-up study (v2): pre-registered, in progress

The diagnostics of §8 suggest specific changes to the quantum model. Testing them honestly needs the same discipline as v1: tune on old data, commit predictions, then test on runs that did not exist when the predictions were written. The study is designed and partly run; its re-test is not finished, and this paper reports none of its results as findings.

Design (`docs/superpowers/specs/2026-09-28--design.md`, committed before any run). The v2 quantum model keeps SMCD’s circuit and adds K copies of it (same encoding scalings, hence the same Ω , each with its own angles) mixed by a linear head, so the head gets independent handles on the Fourier amplitudes that v1’s single readout couples. The boundary ansatz becomes a tuned choice for every family, between v1’s mask ansatz and the affine ansatz of §8. Every classical family receives the same number of tuning configurations as the quantum model. Tuning uses only v1’s Poisson problem with seeds 0–2; the re-test uses seeds 10–14 and re-adjudicates v1’s twelve claims, word for word, with the unchanged script. The selection rule, including a parsimony rule on K , was committed before the first lab result (`scripts/select_v2_config.py`).

What the lab has shown so far (exploratory, not a finding). At the full training budget on Poisson, two copies lower `q_serial`’s median $\text{rel-}L^2$ from 1.63 to 1.27, and the best classical configuration reaches 0.35. A learning rate of 10^{-2} makes the single circuit worse (3.55). The affine-ansatz arm and the larger K are still running. We state now, as the design did, that the v2 model may not beat the classical baselines; the re-test will report whatever it finds.

12 Limitations

Stated plainly, not hedged:

1. **Statistical power, worse than planned.** The core matrix was *pre-registered* at $n = 5$ seeds per (problem, family) cell, which cannot reach $p < 0.05$ on its own (`src/qapinn/stats.py`, T4.1: the $n = 5$ Wilcoxon p -floor is 0.0625). Under the Aug 7 submission deadline, the seed count was cut to $n = 2$ (`docs/plan/08_TASK_INDEX.md`’s cut-line, invoked twice), which lowers the achievable p -floor further, to 0.5 (verified directly against `scipy.stats.wilcoxon`). `docs/predictions.md`’s CONFIRMED gate was left at its original, pre-registered 0.0625 threshold rather than loosened to compensate, which means every ratio-and-significance prediction (PR-1, PR-2, PR-6) can now only resolve to REFUTED or INCONCLUSIVE, independent of true effect size — stated here explicitly rather than left implicit in a table of verdicts.
2. **Simulation only.** All quantum circuits are evaluated on this project’s own statevector simulator (`qsim.py`), cross-checked against a PennyLane oracle to $< 10^{-10}$; no results here were run on physical quantum hardware.
3. **Qubit budget.** Experiments are capped at ≤ 8 qubits, dictated by classical simulation cost, not by the design methodology itself — SMCD’s depth/width formulas (Section 3) apply unchanged at larger n .
4. **Noise models are surrogates.** Shot-noise and depolarizing noise are analytic surrogates (T2.11), not a full density-matrix simulation; the discrepancy between the surrogate and

a density-matrix reference on a representative subset is measured and reported (T4.4), not assumed negligible.

5. **Scope cut.** A fifth PDE (Allen–Cahn / wave, stretch goal P5) was cut from the critical path under the project’s time budget; nothing in Section 3 is specific to the four PDEs actually run.
6. **The affine-encoder constraint (D3).** The classical encoder E_φ is restricted to an affine map, imposed to keep SMCD’s coverage metric well-defined (a nonlinear encoder would make “the circuit’s realized frequency set” itself input-dependent); this is a real restriction on the hybrid model’s expressivity, not a free simplification.
7. **predicted_benefit is a heuristic.** SMCD’s a-priori benefit estimate (Section 3.6’s design card, plotted in `decision_map`) is a provisional heuristic, not a proven bound; Section 5 reports how well it tracks the measured benefit, and a heuristic that tracks poorly is itself a reported finding, not hidden.
8. **PR-10 (barren-plateau decay rate) is unmeasurable on this run’s data, and reported as such rather than computed anyway.** `depth_sweep`’s grid was cut under the same deadline pressure to $n_{\text{qubits}} = 6$ only (dropping the originally planned $n_{\text{qubits}} = 4$ point); `barren_plateau_fit` needs at least two distinct n_{qubits} values to fit a slope, and now detects this degenerate case explicitly (`src/qapinn/xai/gradvar.py`) rather than returning the ill-conditioned solver artifact `numpy.polyfit` would otherwise silently produce against a constant independent variable. The practical frontier (largest n before grad-var collapses below the 10^{-10} floor) is unaffected and still reported; only the decay-rate metric is the casualty.

An infrastructure limitation worth stating explicitly. The core-matrix sweep this project’s results are drawn from ran on a single consumer GPU under real memory and process-stability constraints, and required several crash-recovery restarts over the course of Phase 3. The external retry mechanism resumes correctly by run-completion (`metrics.json` existence), but a known consequence is that a small fraction of runs’ cumulative per-checkpoint diagnostic files (`specerr.npz`) were appended to across a crash boundary, producing duplicate checkpoint entries in that specific artifact (10 of 84 core-matrix runs, detected mechanically by `scripts/check_specerr_integrity.py`). Rather than exclude these runs, the duplicate entries were deduplicated post hoc (`scripts/dedupe_specerr.py`, run once all sweeps had finished so the fix could not race a still-writing task): for each of the affected files, only the last — most recent, post-crash-recovery — row for each real checkpoint step is kept, discarding the stale row written before the crash. This is a mechanical cleanup of an accumulation artifact, not new data; it recovered two figures (`freq_heatmap_poisson`, `freq_heatmap_helmholtz_k4`) and one prediction (PR-8/poisson, from INSUFFICIENT_DATA to a real CONFIRMED; 99.9% overlap after the post-submission correction in §14) that would otherwise have been reported as missing or excluded rather than fixed.

13 Conclusion

Against the contributions claimed in Section 1, adjudicated strictly against the pre-registered falsifiers of `docs/predictions.md` (12 predictions, 15 checks counting per-problem splits; §5–6):

C1 (SMCD’s constructive map) is not supported at the achieved sample size, with one qualified exception at high wavenumber. Its own falsifier, PR-6, is INCONCLUSIVE

on both checked problems — $n = 2$ seeds cannot statistically separate `q_serial` from `q_random`, even where the raw ratio points the predicted way (heat, $12.2\times$). The direct comparison against classical baselines on P1 (PR-1/PR-2/PR-3) shows `q_serial` *underperforming* `c_mlp`, `c_ff`, and `c_rff_matched` alike. Generalizing PR-3 to all six instances (§5) sharpens this: REFUTED (classical decisively wins) on four of six, but genuinely CONFIRMED — matched-capacity statistical equivalence — on the two highest-wavenumber instances, Helmholtz_k10 and Helmholtz_k20, consistent with PR-5’s k -scaling trend (though there every family, classical and quantum, scores $\text{rel-}L^2 \approx 1$: §14, E5). Read together, the honest statement is not “SMCD doesn’t work” but “this run finds no evidence of a quantum *advantage* anywhere, and finds the quantum *disadvantage* narrowing to statistical parity specifically at the hardest, highest-wavenumber instances” — a real, qualified finding, not the clean constructive win C1 was written to claim.

C2 (quantum-NTK block decomposition, flattened decay inside Ω) is reversed where measurable. PR-7 REFUTED on Helmholtz_k10: the quantum block’s spectrum was steeper, not flatter, inside the encoded band on this data. Poisson is INSUFFICIENT_DATA, not REFUTED — fewer than 2 positive eigenvalues fall in the outside-band index range of this run’s committed NTK spectrum, so the comparison this prediction needs is undefined there, reported as such rather than guessed at.

C3 (spectral-bias relief as mechanism) holds on one instance, not the other. PR-8 CONFIRMED on Poisson (99.9% of the real, positive improvement lands inside Ω ; 71.3% in the submitted version, §14) but REFUTED on Helmholtz_k10, where `q_serial` does not beat `c_mlp` at any frequency at all — there is no improvement for the mechanism to explain on the harder instance.

C4 (the negative-result map) is right about the mechanism, wrong about the uniform threshold. Three of four quantum families tested on P2 (heat) underperform `c_mlp` exactly as the smoothing-operator argument predicts, `q_random` by an order of magnitude; `q_serial`’s own modest +7.7% is the one quantum counter-example, and two classical-family violations (`c_ff`, `c_rff_matched`) are unrelated to the quantum-specific claim the 5% bar was meant to test. The mechanistic story — smoothing operators give a high-frequency-rich circuit nothing to earn its keep on — is better supported by the per-family breakdown than by the single uniform threshold PR-4 was pre-registered against.

C5 (the XAI protocol) is the contribution this run actually demonstrates cleanly. Every number above is a falsifiable, mechanically-checked quantity, run identically across classical and quantum families, computed from committed artifacts by `scripts/adjudicate_predictions.py` rather than asserted — and the protocol caught real bugs in its own supporting tooling along the way (a NaN silently miscategorized as REFUTED in PR-8’s early implementation, a degenerate least-squares fit that would have reported a fabricated PR-10 slope, corrupted checkpoint data recovered rather than silently used), which is itself evidence the falsification discipline this paper argues for is not decorative.

The honest headline, matching Section 1’s own framing: at the sample size this deadline allowed ($n = 2$, not the pre-registered $n = 5$), this run does not demonstrate the quantum-advantage hypothesis SMCD was built to test — on the one problem with a full ablation (Poisson), the hybrid model underperforms every classical comparator, and the NTK-block and spectral-flattening mechanisms proposed to explain *why* it should help (C2) point the opposite direction from predicted. What does hold up is the negative-result mechanism on the smoothing control (C4, mostly), one genuine per-frequency win (C3, Poisson only), and the measurement protocol itself (C5) — which is precisely the outcome this project’s own pre-registration discipline exists to make legible rather than obscure. A larger, unrushed replication at the originally planned $n = 5$ seeds, and a `depth_sweep` grid that retains n_{qubits} variation (needed to even attempt PR-10), are the two

most direct next steps (Section 12).

14 Post-submission errata (2026-09-25)

Added after the WISER 2026 BQP Challenge; the version that was judged is repository tag `v1.0-submission`. The pre-registered thresholds and all 15 verdicts are unchanged. Every number below was re-derived from the committed runs; `FINDINGS.md` gives the reproduction commands. E1–E3 are corrections, E4–E6 caveats the results understate.

- E1 (PR-7).** The `Helmholtz_k10` decay-exponent gap is -8.37 , not -9.67 . The old value came from reloading gitignored checkpoints, before PR-7 was switched to the committed `xai/ntk_step*.npz`. Still REFUTED.
- E2 (PR-8).** 99.9%, not 71.3%, of Poisson’s improvement lies inside Ω . Poisson’s error spectrum is two-sided and $\Omega = -\Omega$, but the overlap code folded Ω onto $|\omega|$. So the mirrored half of the improvement (3,944 of 13,751 units; 5,864 sit at $\omega = 0$) counted as outside Ω . Still CONFIRMED.
- E3 (figures).** The submitted heatmaps and `ntk_spectrum_p1/p4` shaded the design card’s four encoding scalings ($\pi, 3\pi, 9\pi, 27\pi$) as Ω instead of the realised frequency set (81 frequencies for Poisson). Adjudication always used the correct Ω ; the figures in this revision are regenerated.
- E4 (PR-9).** The coverage sweep trained at 1500+150 steps. Its six targets build only three distinct Poisson circuits: 0.4 and 0.6 give one, 0.8, 0.9 and 1.0 give another, and 0.2 gives a third. The 18 points are therefore 9 distinct measurements, and coverage is confounded with parameter count (8 vs. 14). At this budget full coverage wins on every seed (0.718–0.729 vs. 0.909–1.084). At the full 20000+2000 budget the order reverses on two of three seeds: 1-layer circuits reach 0.66–0.70, while the 4-layer circuit stays at 1.54–1.58. `Helmholtz_k10` builds a single circuit at every target and scores 0.991–1.002 throughout.
- E5 (PR-3, PR-5).** At $k = 10$ and $k = 20$ every family’s median $\text{rel-}L^2$ is 0.977–1.089, while predicting $u \equiv 0$ scores exactly 1. The CONFIRMED parity there is shared failure, not matched capability. SMCD’s own predicted benefit is 0 at $k = 4$ and $k = 10$.
- E6 (PR-1, PR-2).** The core matrix trained each family at its default size. On Poisson `q_serial` has 14 parameters, `c_mlp` 8,513 and `c_ff` 129. Only `c_rff_matched` (13), `q_random` (14) and `q_octave` (13) are size-matched.

15 Reproducibility appendix

Every number below is measured against this session’s actual runs, not estimated.

- **Hardware:** NVIDIA RTX 4090 Laptop GPU, 32 logical CPU cores (`provenance.json`, recorded per run) — one machine for every run in this paper.
- **Software** (`docs/ENV_RESOLVED.md`, pinned via `uv`): Python 3.12.13, `torch` 2.11.0+cu128 (CUDA 12.8), `pennylane` 0.45.1, `numpy` 2.4.4, `scipy` 1.18.0.

- **Determinism:** `qapinn.seeding.set_global_seed` seeds Python, NumPy, and Torch (CPU+CUDA) from one integer and enables `torch.use_deterministic_algorithms`; an explicit `torch.Generator` is threaded through every sampling call, so two runs at the same seed on the same device produce bit-identical `metrics.json` (T0.20). A CPU run and a CUDA run of the same configuration round differently (by 10^{-15} in $\text{rel-}L^2$ on a smoke run we re-checked; training can amplify such differences), and the run ID does not include the device.
- **Seeds used** (state explicitly, cut from a pre-registered $n = 5$ under the Aug 7 deadline — Section 12, FINDINGS.md F10): `core_matrix` $\{0,1\}$; `coverage_sweep/alpha_sweep/noise_study/soft_bc_ntk` $\{0,1,2\}$; `depth_sweep` $\{0\}$.
- **Concurrency:** `--workers 2 --mem-fraction 0.35` for every sweep except `depth_sweep` (`--workers 1 --mem-fraction 0.8`, forced down after system-memory exhaustion on its larger circuits — OS/driver contention under concurrent CUDA context creation on this specific machine, not a VRAM limit).
- **Measured wall-clock:** `tasks.py repro-quick` (CPU-only, 42 combinations, implemented this session): **513.7s (≈ 8.6 min)**, 42/42 OK, under the 15-min target. `tasks.py test` (385 tests, post-submission revision): 244s CPU-only. `tasks.py figures` (20 figures from committed data, no GPU, byte-identical on the same machine): about 30s. `tasks.py repro-all` (added after submission): re-derives every figure and verdict from the committed runs in 56s; retraining all 197 runs from scratch is 67.2h of training wall-clock on this machine (the sum of the runs’ recorded `wall_clock_s`; `core_matrix` alone is 58.5h).

References

- [1] Amira Abbas, David Sutter, Christa Zoufal, Aurelien Lucchi, Alessio Figalli, and Stefan Woerner. The power of quantum neural networks. *Nature Computational Science*, 1:403–409, 2021.
- [2] Ziv Chen, Gal G. Shaviner, Hemanth Chandravamsi, Shimon Pisnoy, Steven H. Frankel, and Uzi Pereg. Quantum physics-informed neural networks for Maxwell’s equations: circuit design, “black hole” barren plateaus mitigation, and GPU acceleration. *Quantum Machine Intelligence*, 8:21, 2026. doi: 10.1007/s42484-026-00365-w. arXiv:2506.23246.
- [3] Salah A. Faroughi and Farinaz Mostajeran. Neural tangent kernel analysis to probe convergence in physics-informed neural solvers: PIKANs vs. PINNs. arXiv:2506.07958, 2025.
- [4] R. Allan Freeze and John A. Cherry. *Groundwater*. Prentice-Hall, Englewood Cliffs, NJ, 1979.
- [5] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, 2018.
- [6] Said Lantigua, Gilson Giraldo, and Renato Portugal. A classical-quantum hybrid architecture for physics-informed neural networks. arXiv:2511.07216, 2025.
- [7] Jarrod R McClean, Sergio Boixo, Vadim N Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9: 4812, 2018.

- [8] M. Z. Naser. Fundamental flaws of physics-informed neural networks and explainability methods in engineering systems. *Computers & Industrial Engineering*, 212:111704, 2026. doi: 10.1016/j.cie.2025.111704.
- [9] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [10] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. The effect of data encoding on the expressive power of variational quantum machine learning models. *Physical Review A*, 103(3): 032430, 2021.
- [11] Alexandr Sedykh, Maninadh Podapaka, Asel Sagingalieva, Karan Pinto, Markus Pflitsch, and Alexey Melnikov. Hybrid quantum physics-informed neural networks for simulating computational fluid dynamics in complex shapes. *Machine Learning: Science and Technology*, 5(2): 025045, 2024. doi: 10.1088/2632-2153/ad43b2.
- [12] Matthew Tancik et al. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, 2020.
- [13] David Keith Todd and Larry W. Mays. *Groundwater Hydrology*. John Wiley & Sons, Hoboken, NJ, 3 edition, 2005.
- [14] Kaveh Zabihi, Hamid Montazeri, and Akke S. J. Suiker. Hybrid quantum-classical physics-informed neural networks for solving nonlinear PDEs: when and where hybridization is effective? arXiv:2606.04679, 2026.

A Figure gallery

Every figure in this appendix is drawn from committed run artifacts by `scripts/report/` (no retraining) and is also in `results/report/figures/`. Colours follow one fixed family palette, checked for colour-vision deficiency; families are also distinguished by marker and position.

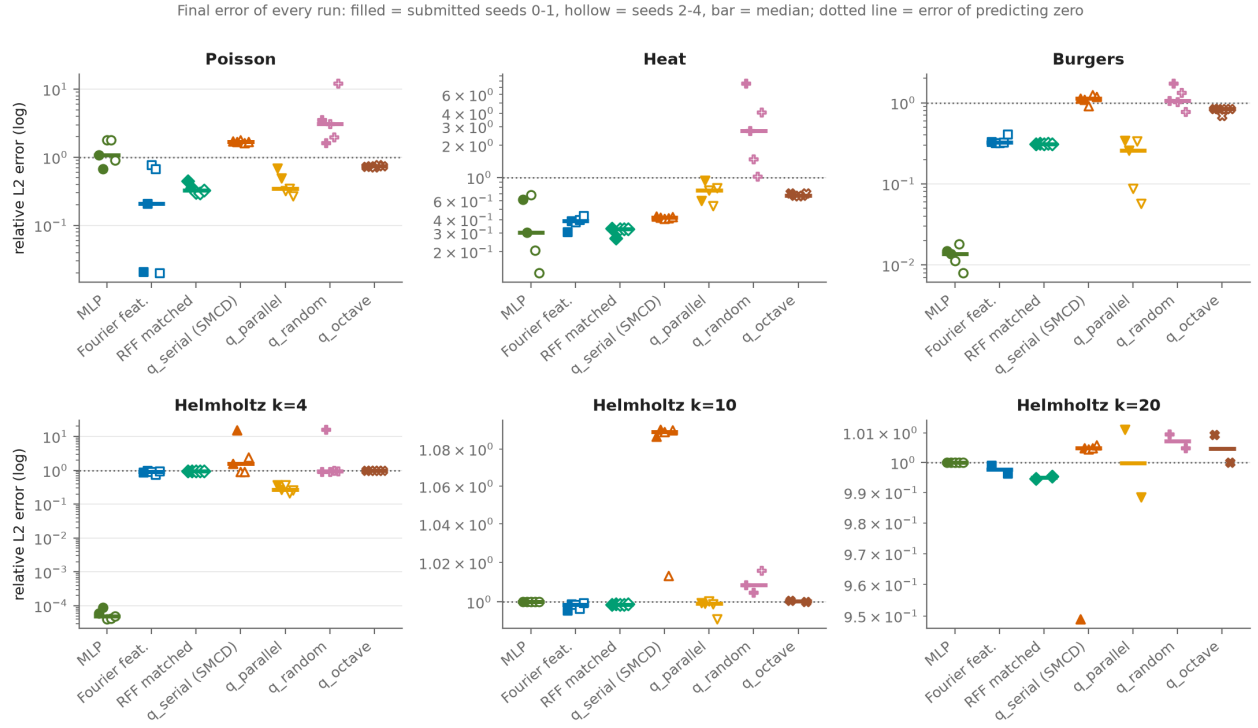


Figure 14: Final relative L^2 error of every core-matrix run at the pre-registered five seeds. Filled: seeds 0–1 (submitted); hollow: seeds 2–4 (post-submission); bar: median; dotted: the error of predicting zero.

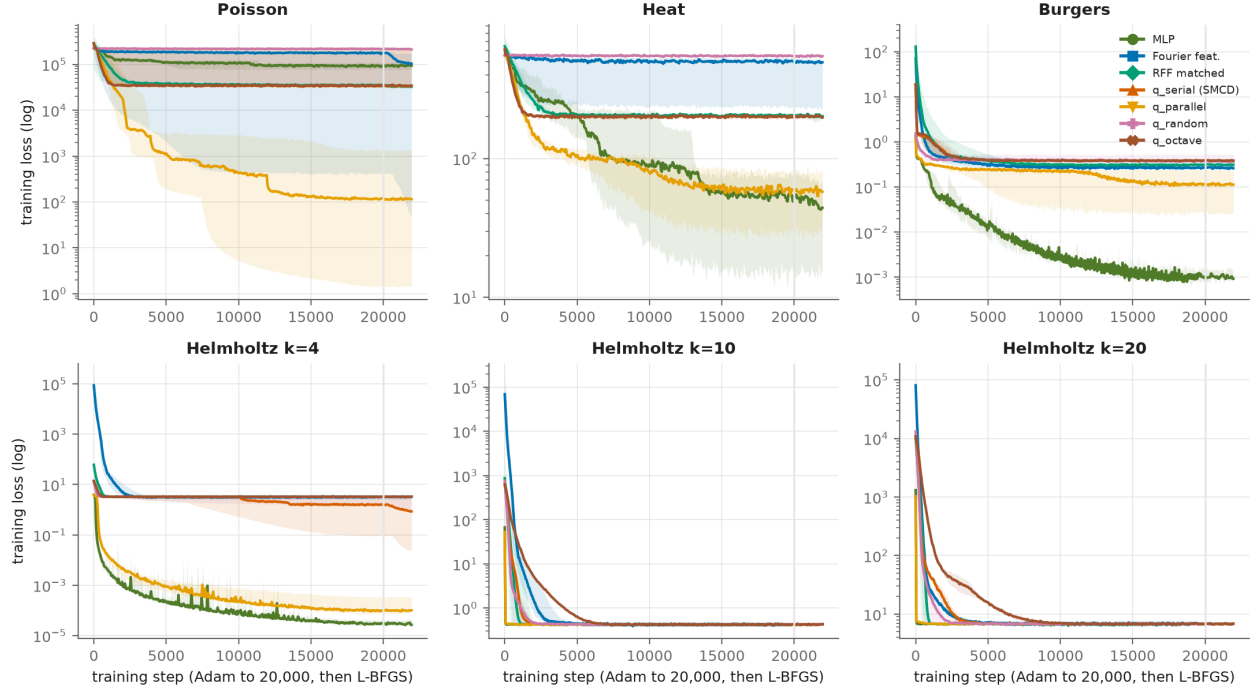


Figure 15: Training loss, median over seeds with the seed range shaded. Adam runs to step 20,000, then L-BFGS.

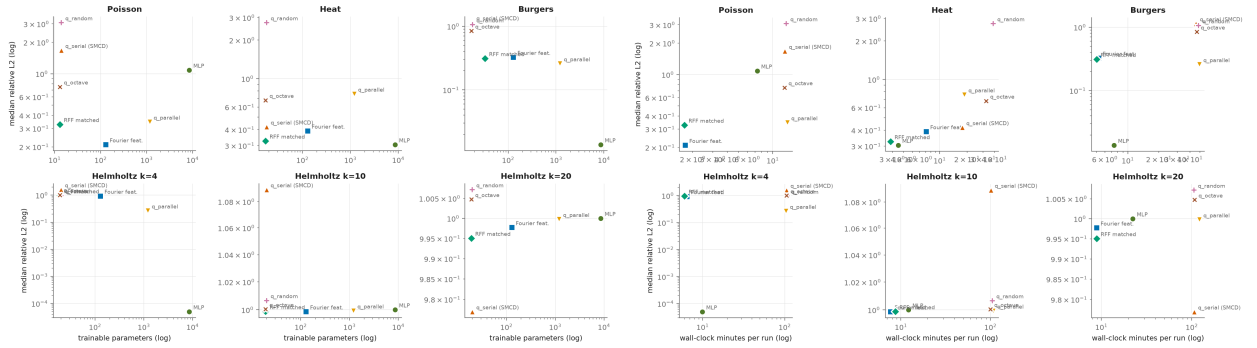


Figure 16: Median error against parameter count (left) and wall-clock time per run (right), per problem.

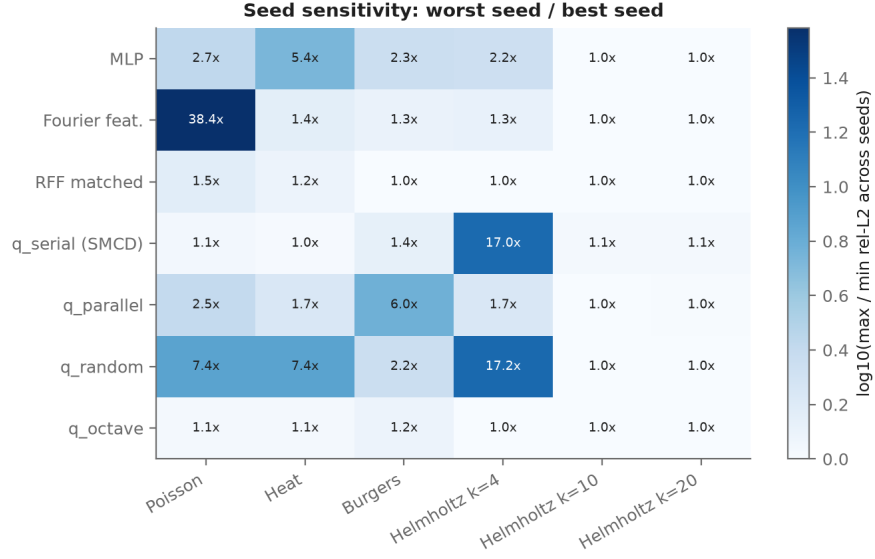


Figure 17: Seed sensitivity: worst-seed over best-seed error for each family and problem.

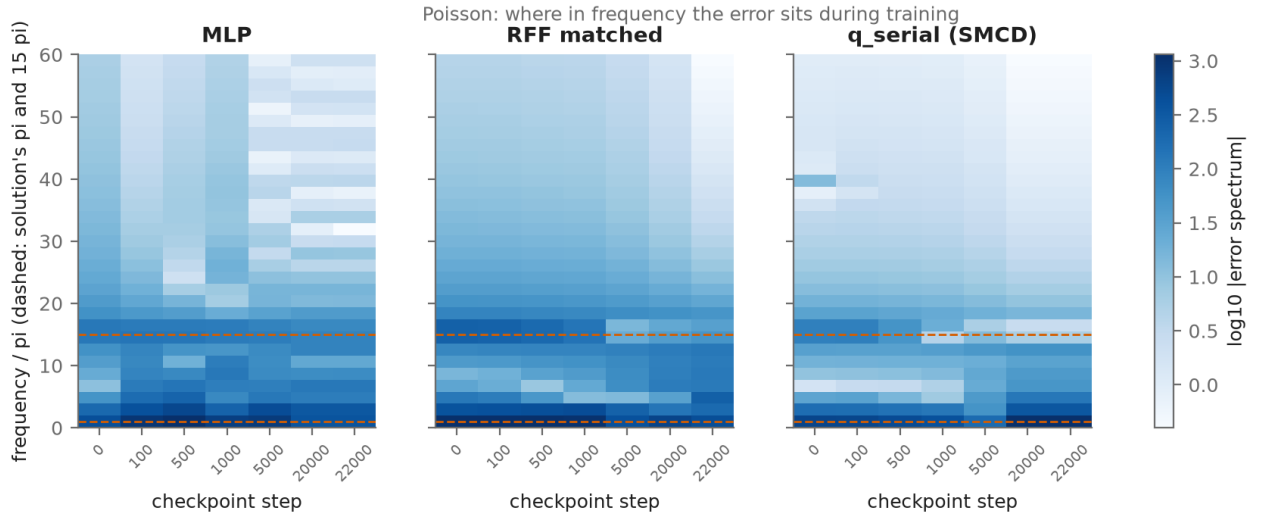


Figure 18: Poisson: magnitude of the error spectrum at each checkpoint (seed 0), with the solution's two frequencies π and 15π dashed. **q_serial** removes the 15π error early and later loses the π component.

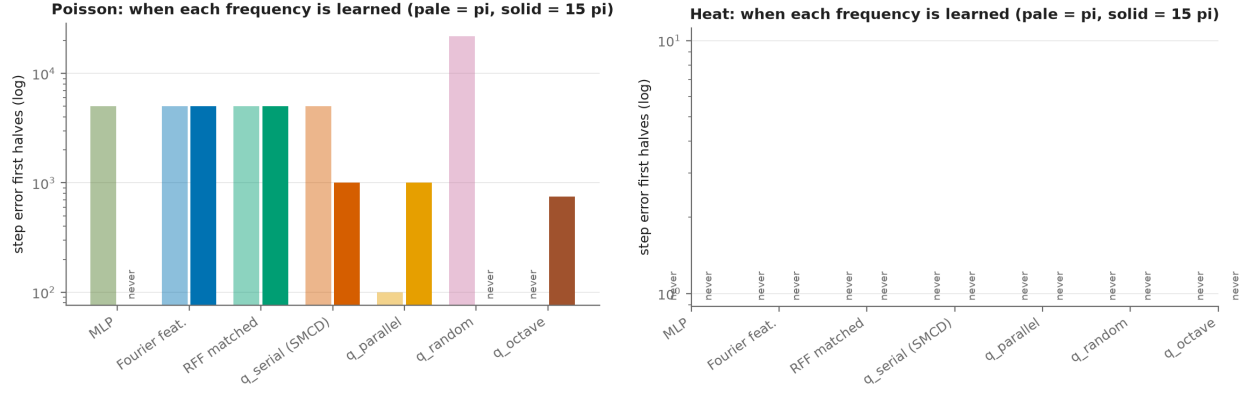


Figure 19: Step at which the error at each solution frequency first halves (median over seeds; pale: π , solid: 15π). Left: Poisson; right: heat.

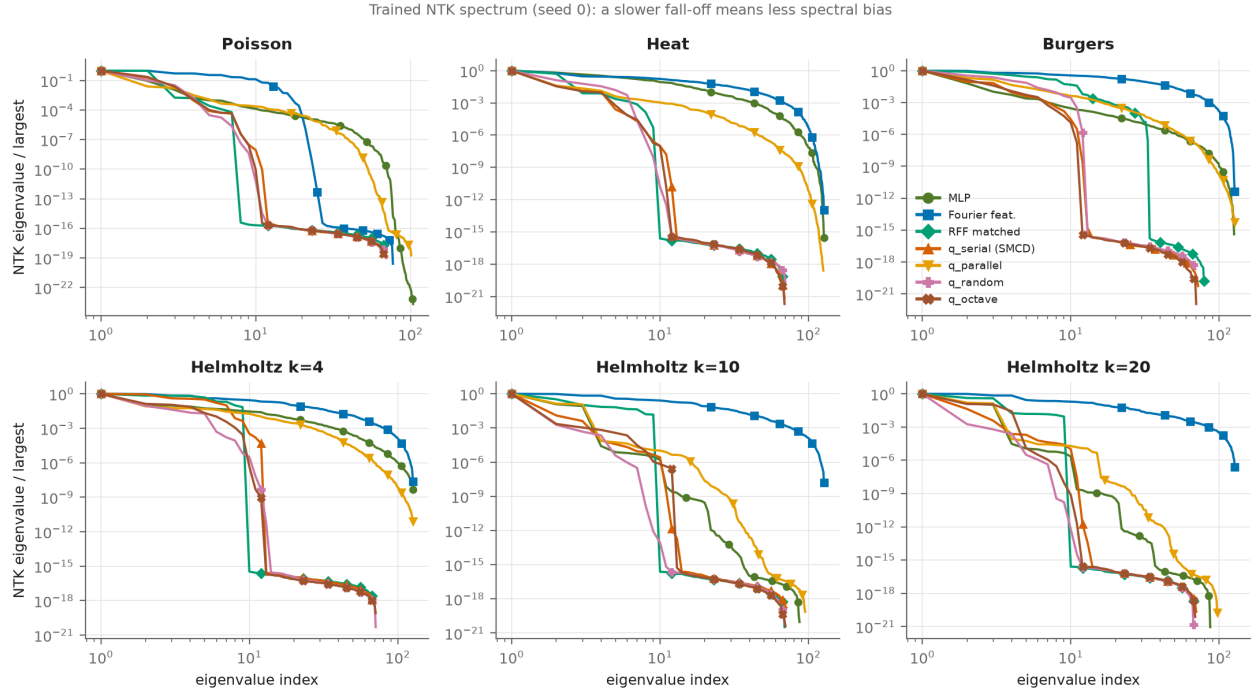


Figure 20: Trained NTK spectra (seed 0), normalised by the largest eigenvalue.

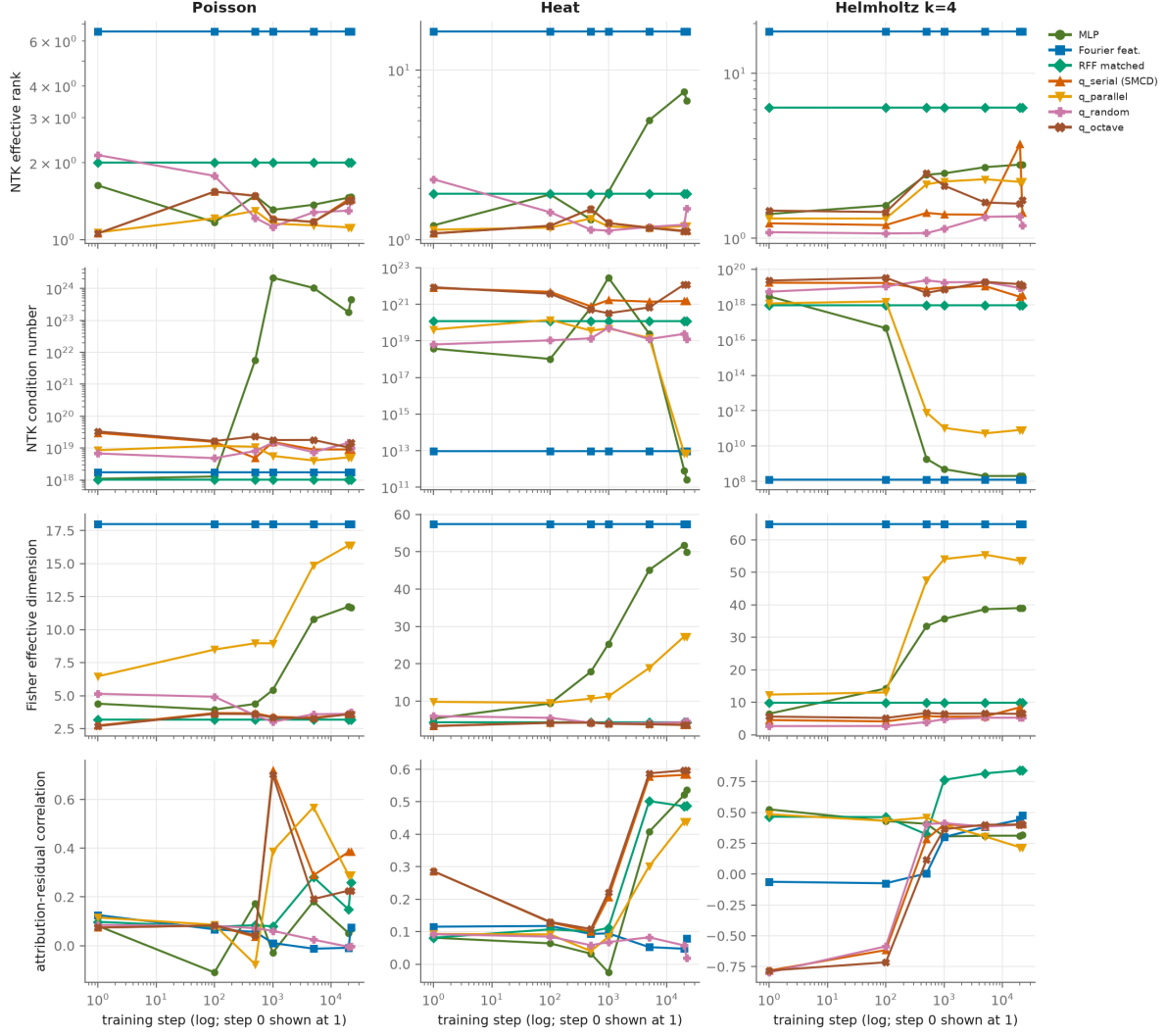


Figure 21: NTK effective rank and condition number, Fisher effective dimension and the correlation of attributions with the residual, over training (median over seeds).

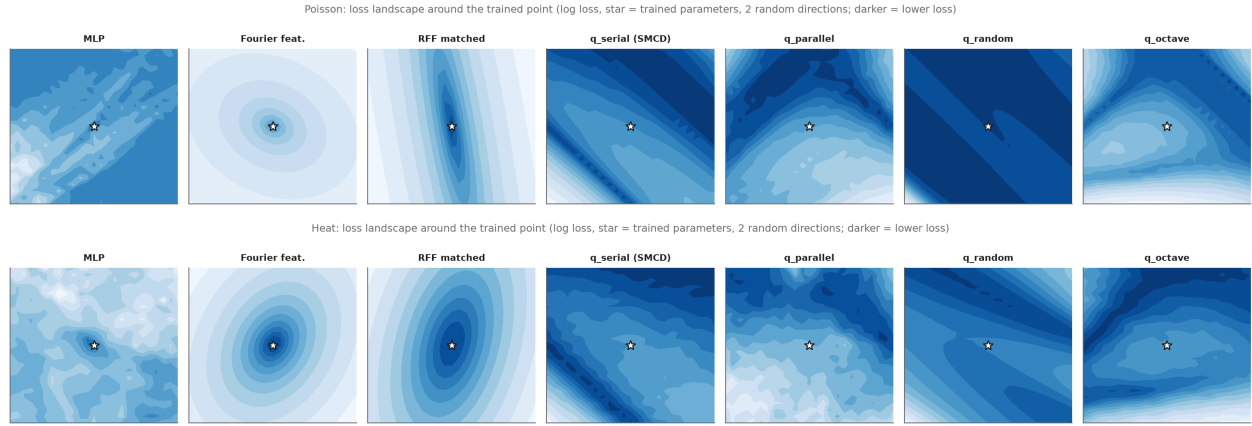


Figure 22: Loss landscape on two random directions around the trained parameters (star), Poisson (top) and heat (bottom); darker is lower loss.

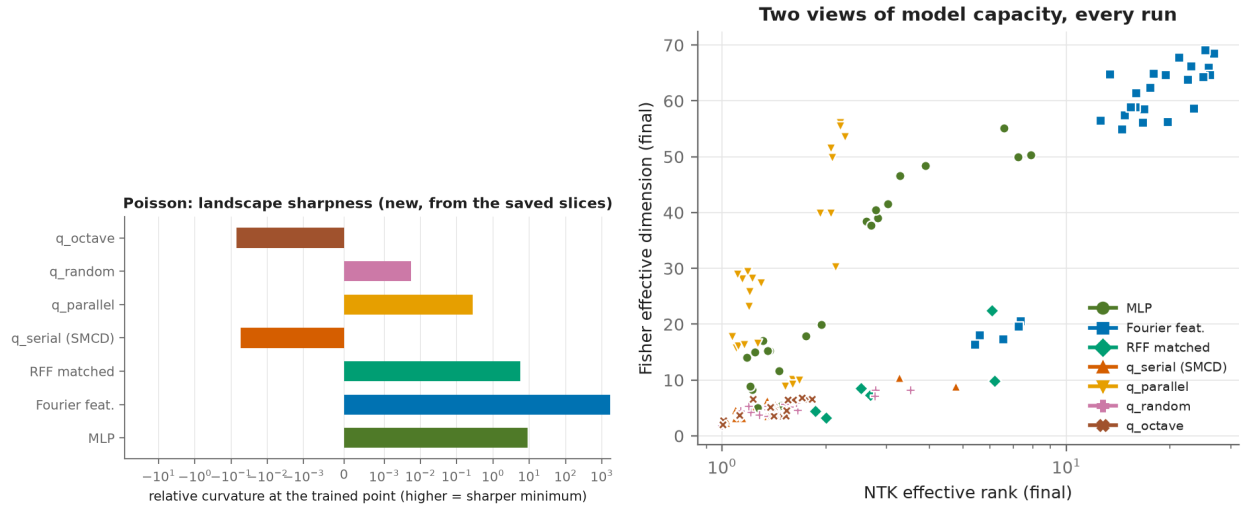


Figure 23: Left: curvature of the loss landscape at the trained point (Poisson). Right: NTK effective rank against Fisher effective dimension, every run.

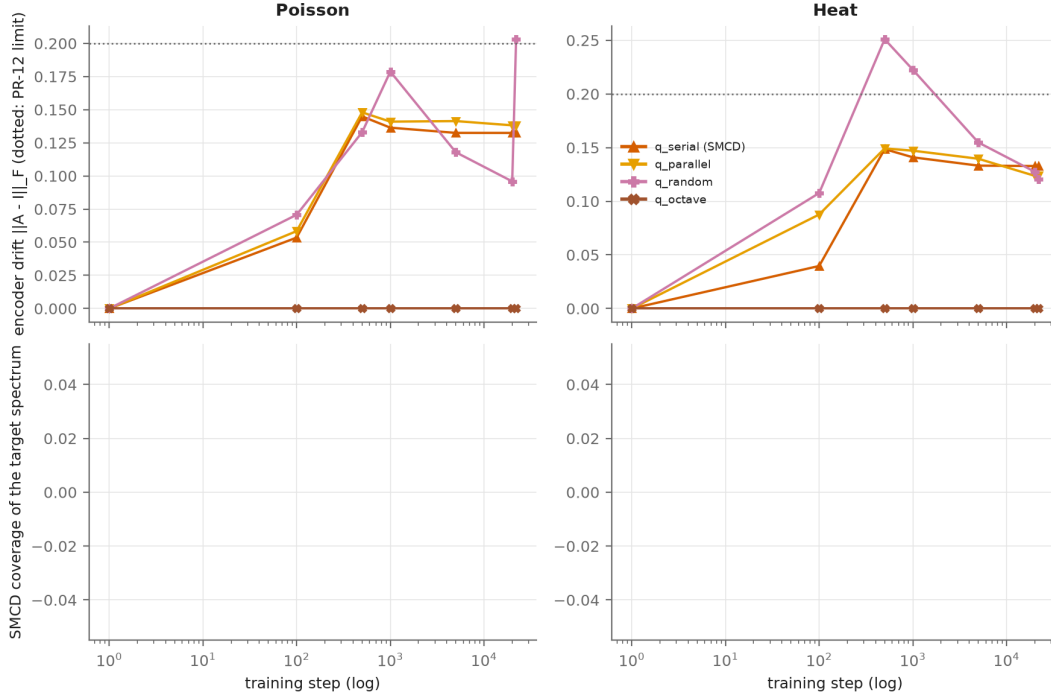


Figure 24: Quantum families: encoder drift $\|A - I\|_F$ (dotted: PR-12's limit 0.2) and SMCD coverage of the target spectrum over training.

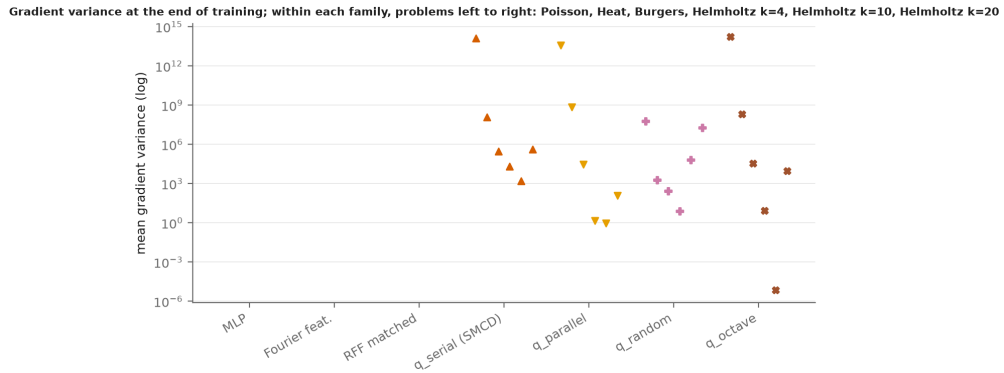


Figure 25: Mean gradient variance at the end of training, per family and problem.

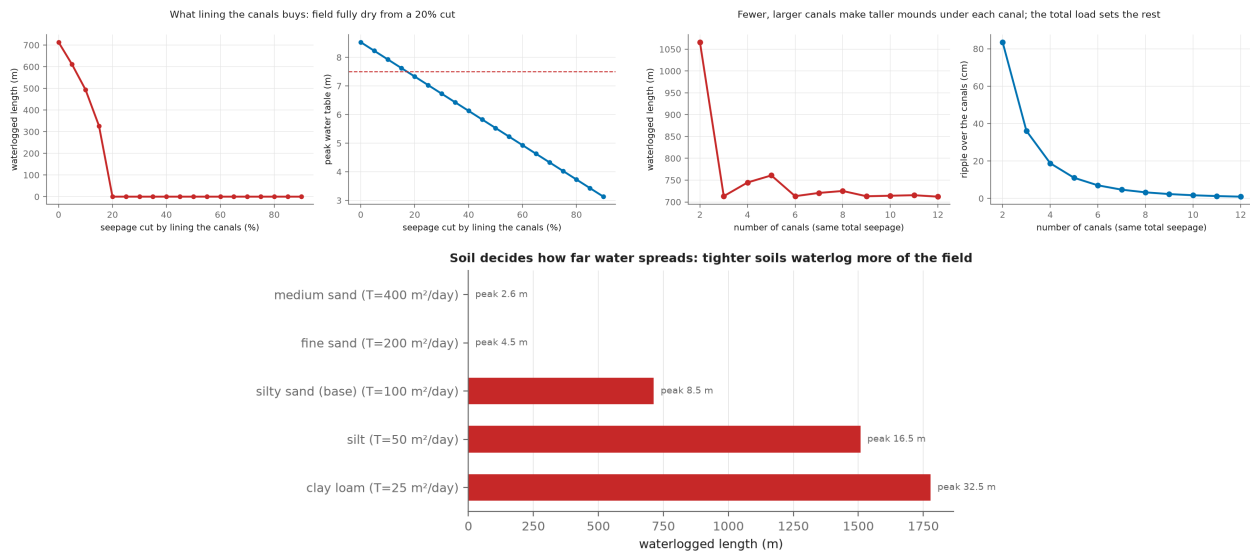


Figure 26: Groundwater scenarios from the exact solution: lining the canals, canal spacing at the same total seepage, and soil type.